Set 2A: Ideal Op-Amp Blocks and Cascades

ECEN 325 — Lecture 2 (Operational Amplifiers) practice set

What this set trains. Circuits built from the standard op-amp blocks (inverting, non-inverting, difference, summing and T-network). With the two ideal op-amp rules (virtual short, no input current) every block reduces to a one-line formula in terms of impedances, and a cascade is the product of its blocks. Problems 1–6 are easy, 7–12 medium, 13–18 hard. All op-amps are ideal. Try each problem before looking at its solution.

The Lecture-2 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1 tools for impedances, dividers, Bode plots and sinusoids):

T1 *Ideal op-amp with negative feedback:* no current into + or −; $V_- = V_+$ (virtual short). Its output is an ideal voltage source (loads on it do not matter).

T2 *Block formulas:* inverting $\frac{V_o}{V_{in}} = -\frac{Z_f}{Z_{in}}$; non-inverting $\frac{V_o}{V_+} = 1 + \frac{Z_f}{Z_g}$ (with V_+ from the divider at the + input); a follower has gain 1. Several blocks in cascade: multiply.

T3 *KCL with impedances:* at a node, $\sum_k \frac{V_{node} - V_k}{Z_k} = 0$ (for a capacitor $\frac{1}{Z} = sC$). Write it at every node except op-amp outputs and source nodes.

T4 *Solving KCL equations:* pick the equation with the fewest unknowns, solve it for one node voltage, substitute into the rest; repeat until only V_o and V_i remain.

T5 *Finite gain:* $V_{out} = A(s)(V_+ - V_-)$, $A(s) = \frac{A_0}{1+s/\omega_b}$, $\omega_t = A_0\omega_b$. With $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$: $\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1+A\beta}$.

T6 *Slew rate:* for $V_{out} = \hat{V}\sin\omega t$, the fastest change is $\left|\frac{dV_{out}}{dt}\right|_{max} = \omega\hat{V}$; no distortion requires $\omega\hat{V} \le SR$.

Steps:

1. *Sources.* DC and AC both present ⇒ superposition.
2. *DC circuit:* $V_i \to 0$, $C \to$ open; pass the DC through each block with its gain at $s = 0$.
3. *Op-amp rules:* confirm negative feedback; apply T1 (or T5 for a non-ideal op-amp).
4. *Transfer function:* standard blocks → T2; anything else → T3 + T4.
5. *Standard form*, 6. *Bode*, 7. *Output* — exactly as in Lecture 1.

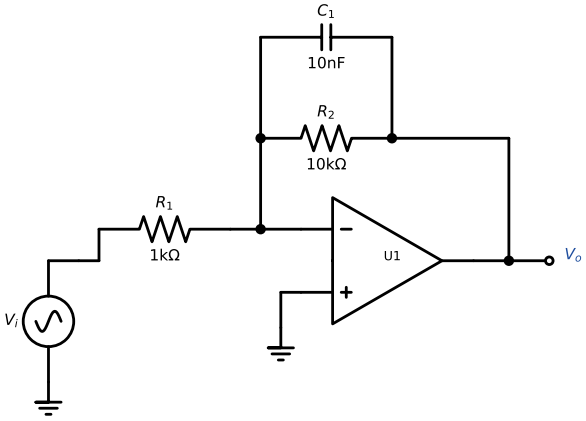
Variations: a design problem runs these steps backwards (Bode plot → $H(s)$ → blocks → component values); a slew-rate problem stops after Step 4 for every op-amp output and applies T6.

Problems

Easy problems

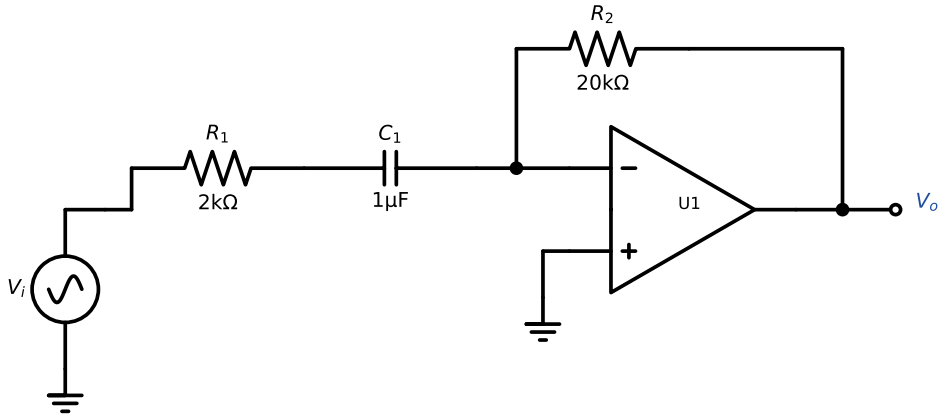
Problem 2A.1 — Easy

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1 \sin(2\pi \cdot 2000\, t)$.



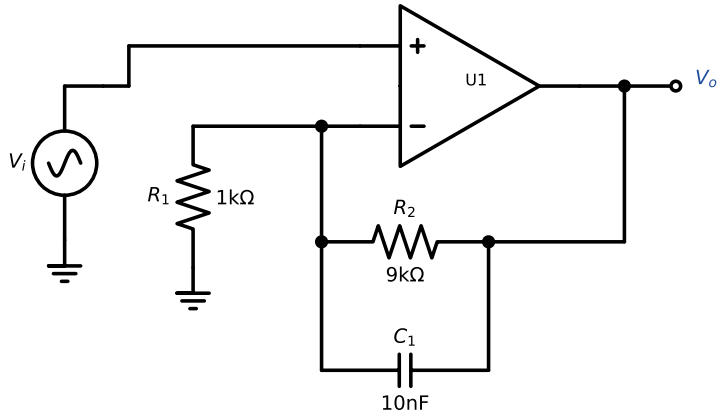
Problem 2A.2 — Easy

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2 \sin(2\pi \cdot 50\, t)$.



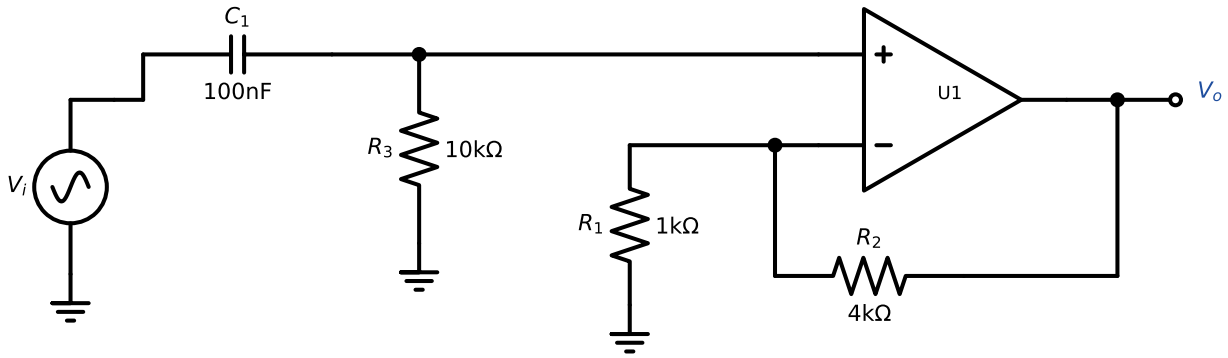
Problem 2A.3 — Easy

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 5000\,t)$.



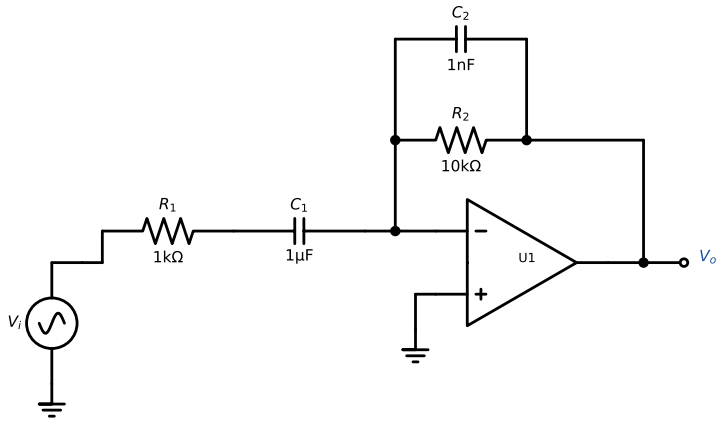
Problem 2A.4 — Easy

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 100\,t)$.



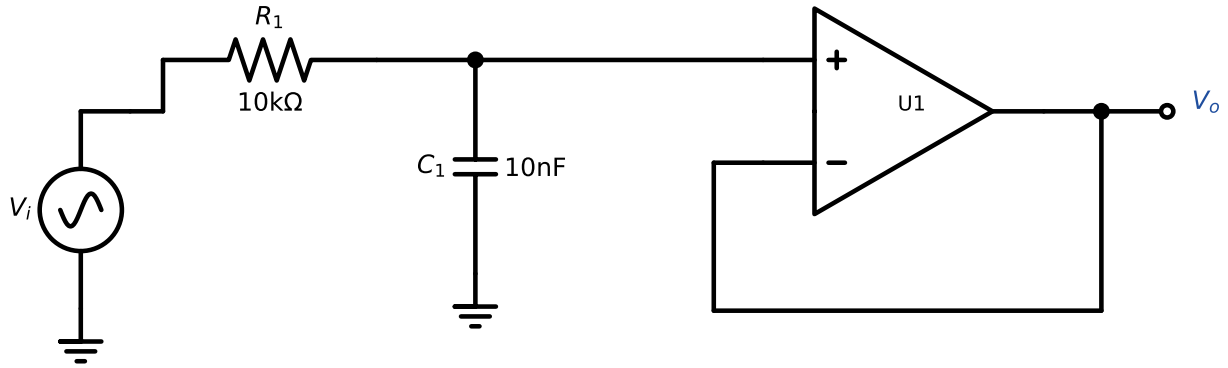
Problem 2A.5 — Easy

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05 \sin(2\pi \cdot 1000\, t)$.



Problem 2A.6 — Easy

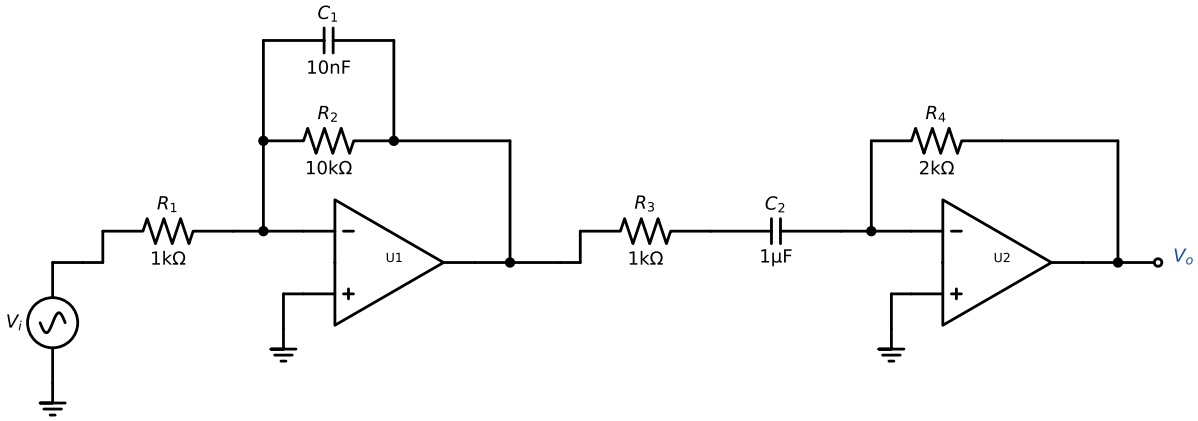
For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 1 \cos(2\pi \cdot 3000\, t)$.



Medium problems

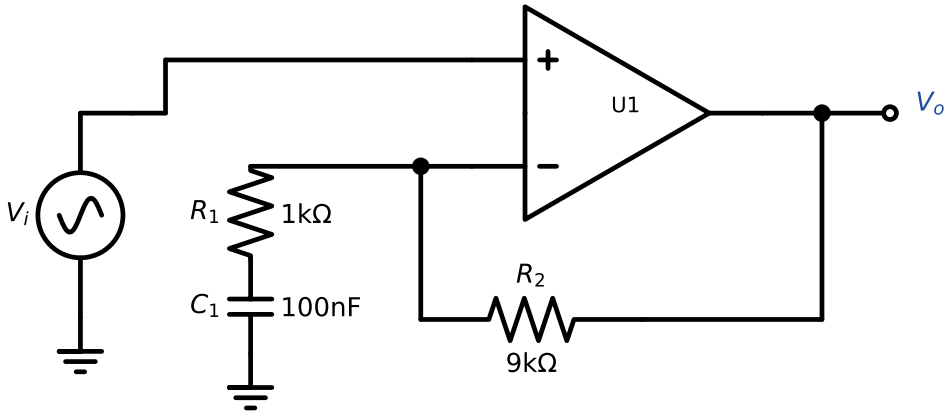
Problem 2A.7 — Medium

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.02\sin(2\pi \cdot 500\,t)$.



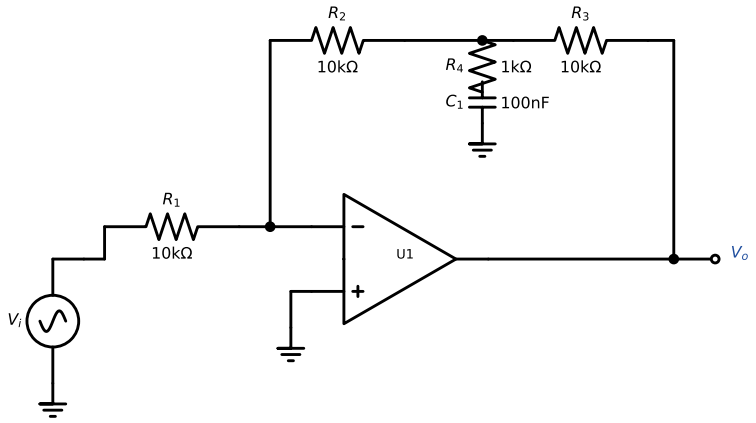
Problem 2A.8 — Medium

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 500\,t)$.



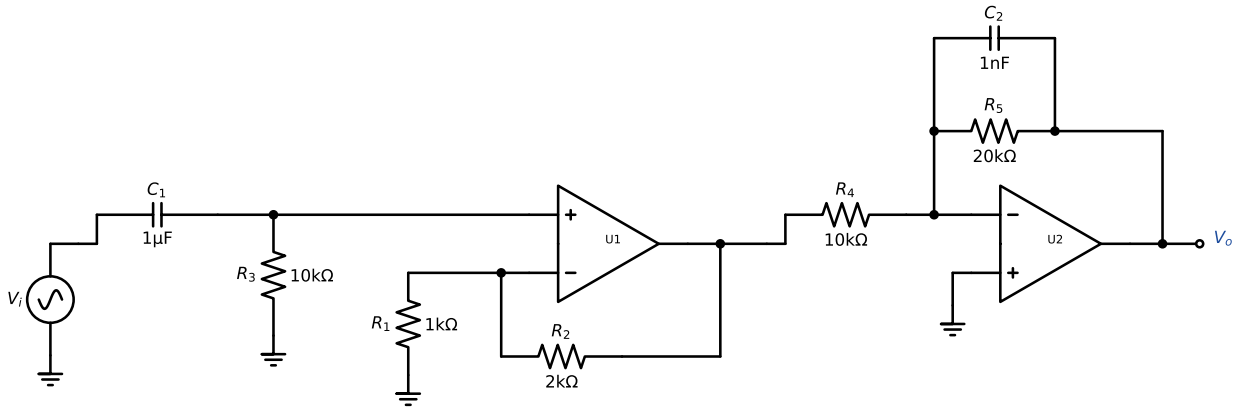
Problem 2A.9 — Medium

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 1000\,t)$.



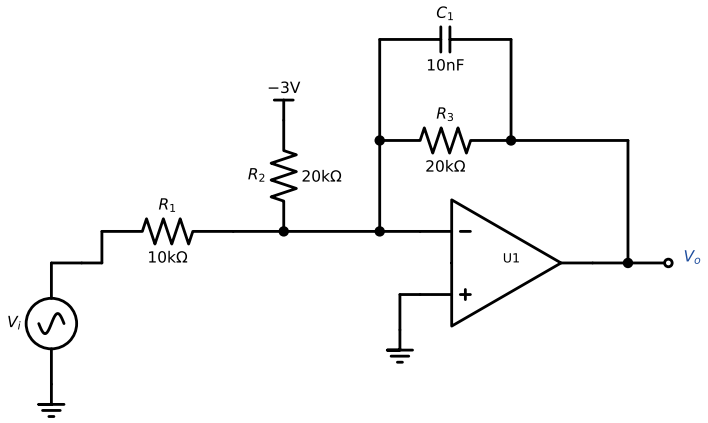
Problem 2A.10 — Medium

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 2000\,t)$.



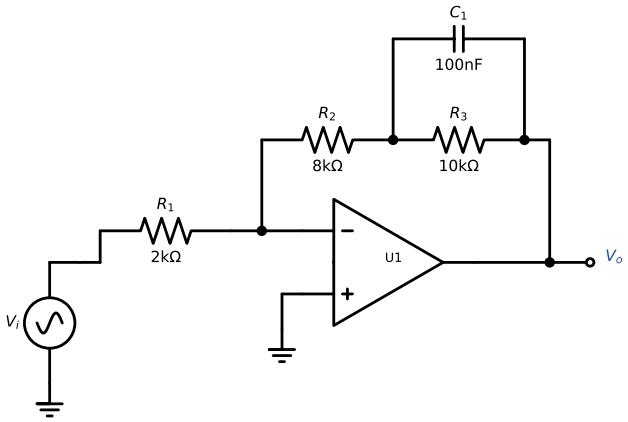
Problem 2A.11 — Medium

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 1000\,t)$.



Problem 2A.12 — Medium

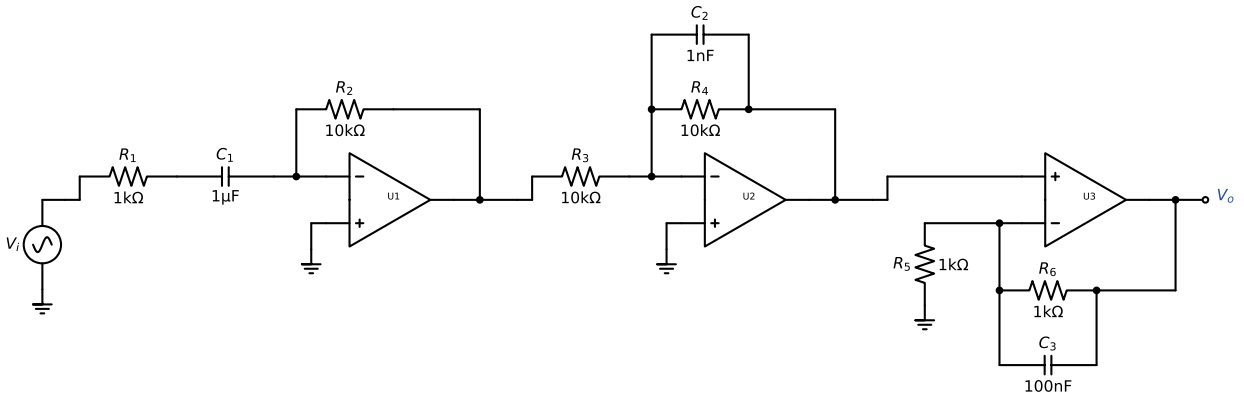
For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 300\,t)$.



Hard problems

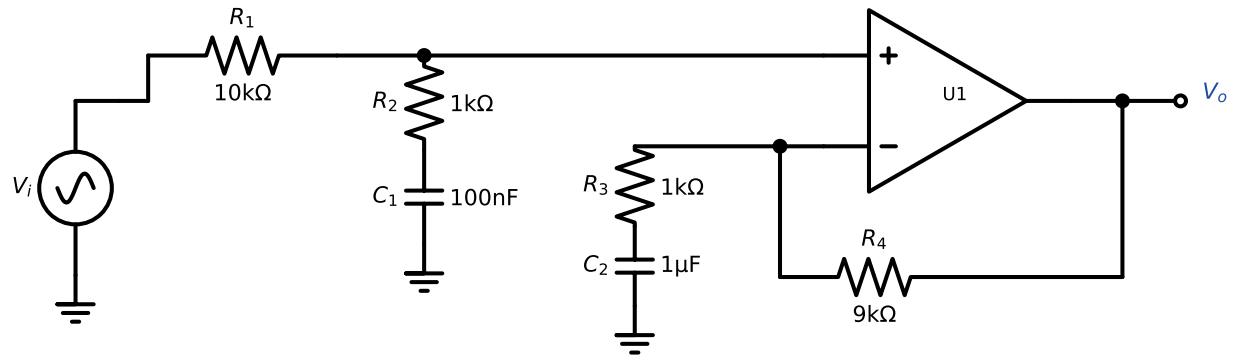
Problem 2A.13 — Hard

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.01 \sin(2\pi \cdot 2000\, t)$.



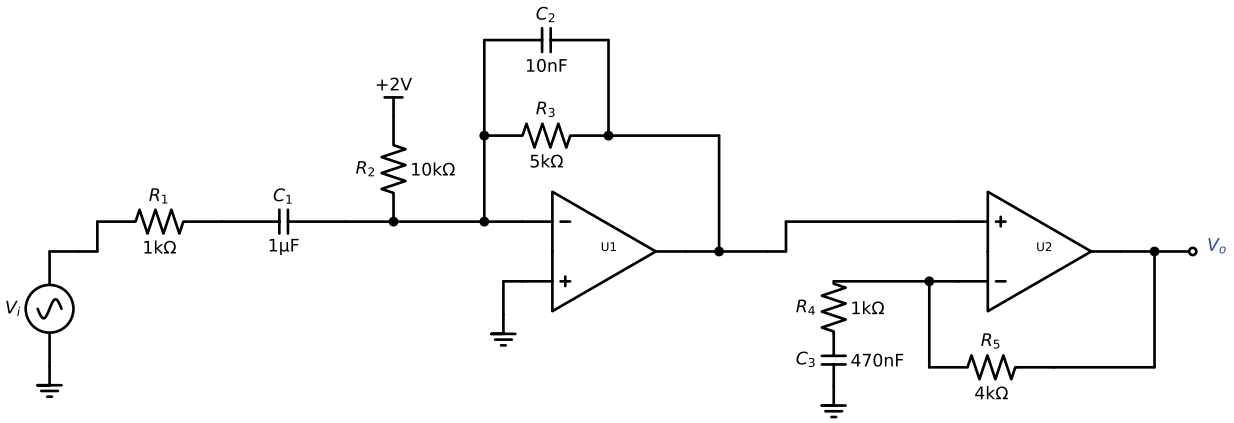
Problem 2A.14 — Hard

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1 \sin\left(2\pi \cdot 1000\, t + \frac{\pi}{4}\right)$.



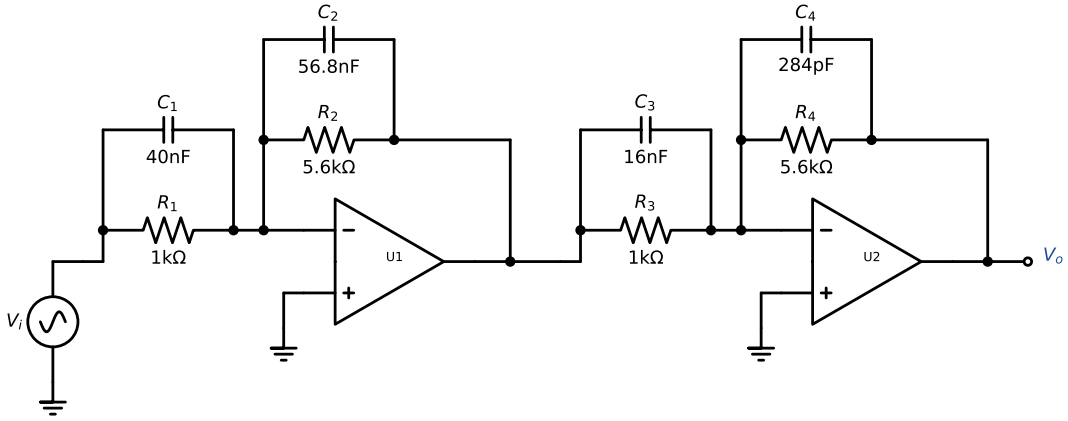
Problem 2A.15 — Hard

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05\sin(2\pi \cdot 1000\,t)$. (The DC supply in the first block is $+2$ V.)



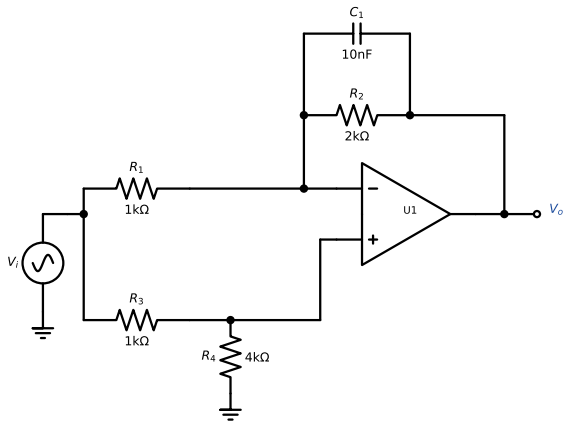
Problem 2A.16 — Hard

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.01\sin(2\pi \cdot 2000\,t)$.



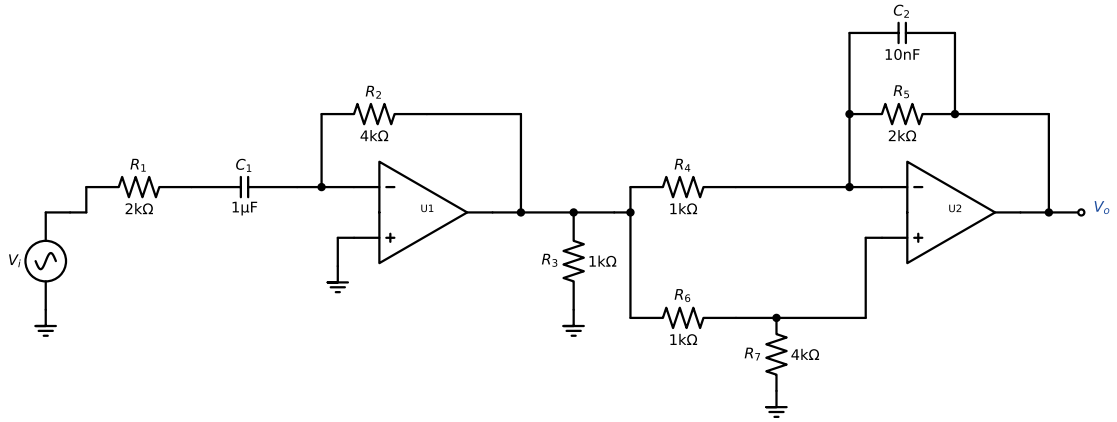
Problem 2A.17 — Hard

For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\cos\left(2\pi \cdot 10000\,t - \frac{\pi}{3}\right)$.



Problem 2A.18 — Hard

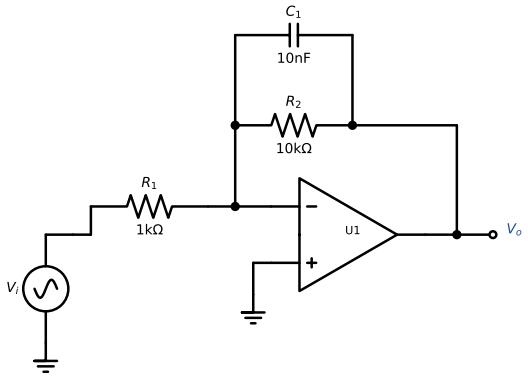
For the circuit below (ideal op-amps): (a) find $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin\left(2\pi \cdot 3000\,t + \frac{\pi}{6}\right)$.



Solutions

Every solution follows the steps of the Lecture-2 pipeline card.

Solution 2A.1 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

The op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a standard block, so use its block formula (tool T2).

Step 4 — Transfer function, block by block

Block 1 (U1). Inverting block: the $+$ input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{R_1}$ through R_1 cannot enter the op-amp, so all of it flows through Z_{f1}: $V_o = 0 - Z_{f1} \cdot \frac{V_i}{R_1}$.

$$Z_{f1} = R_2 \parallel \frac{1}{sC_1} = \frac{10^8}{s + 10000}$$

$$\frac{V_o}{V_i} = -\frac{Z_{f1}}{R_1} = \frac{-10^5}{s + 10000}$$

So:

$$H(s) = \frac{-10^5}{s + 10000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 10000$; $s + 10000 = 0 \Rightarrow s = -10000$.

Factored form (leading coefficient in front):

$$H(s) = -10^5 \, \frac{1}{(s + 10000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -10\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}, \qquad K = -10$$

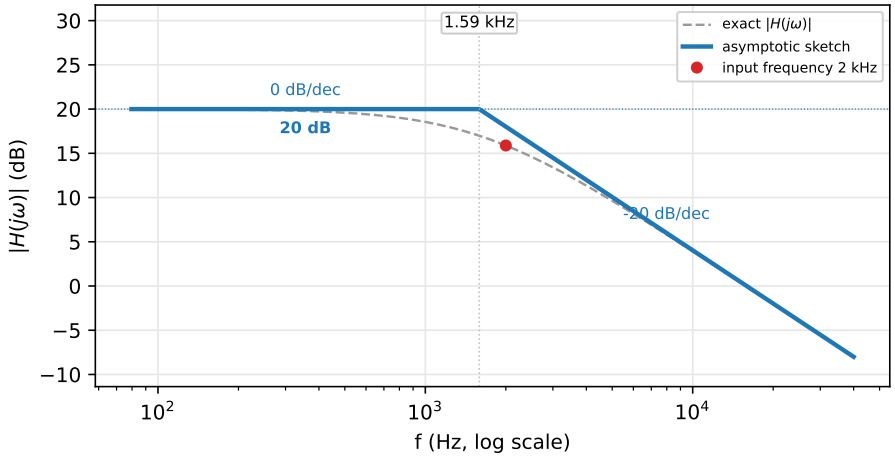
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	10000	1.59 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 10 = 20\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.59$ kHz	0 dB/dec (flat)	20 dB at end
$f > 1.59$ kHz	-20 dB/dec	20 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 10 = 20\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi\cdot 2000\,t)$, so $\omega_{in} = 2\pi\cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-10	10	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}$	0.6227	-0.899
total	**6.227** (product)	$-$**4.04** (sum)

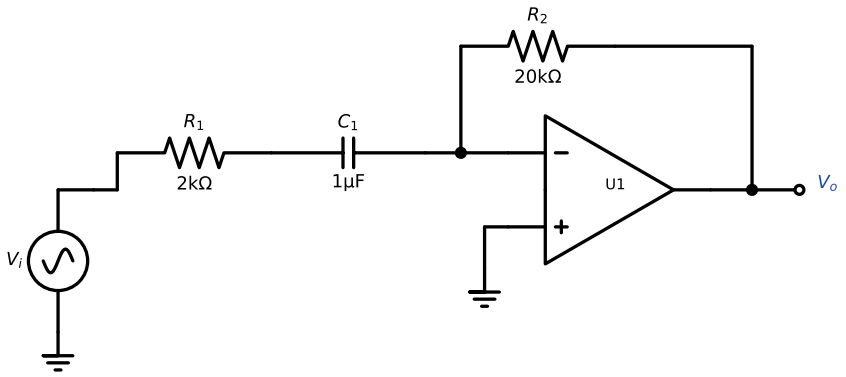
$$|H(j\omega_{in})| = 6.23, \qquad \angle H(j\omega_{in}) = -4.04 \text{ rad}$$
$$v_{o,ac}(t) = 6.23 \times 0.1 \sin(2\pi\cdot 2000\,t - 4.04) = 0.623 \sin(2\pi\cdot 2000\,t - 4.04)$$

$$\boxed{V_o(t) = 0.623 \sin(2\pi\cdot 2000\,t - 4.04) \text{ V}}$$

Answer

$$H(s) = -10\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}} \quad \text{(corners: 1.59 kHz)}$$
$$V_o(t) = 0.623 \sin(2\pi\cdot 2000\,t - 4.04) \text{ V}$$

Solution 2A.2 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

The op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a standard block, so use its block formula (tool T2).

Step 4 — Transfer function, block by block

Block 1 (U1). Inverting block: the $+$ input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{Z_{in1}}$ through Z_{in1} cannot enter the op-amp, so all of it flows through R_2: $V_o = 0 - R_2 \cdot \frac{V_i}{Z_{in1}}$.

$$Z_{in1} = R_1 + \frac{1}{sC_1} = \frac{2000\,s + 10^6}{s}$$

$$\frac{V_o}{V_i} = -\frac{R_2}{Z_{in1}} = \frac{-10\,s}{s + 500}$$

So:

$$H(s) = \frac{-10\,s}{s + 500}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow$ 1 zero at $s = 0$.
- Denominator: $s + 500$; $s + 500 = 0 \Rightarrow s = -500$.

Factored form (leading coefficient in front):

$$H(s) = -10\,\frac{s}{(s + 500)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -10\,\frac{s}{s + 2\pi \cdot 79.6}, \qquad K = -10$$

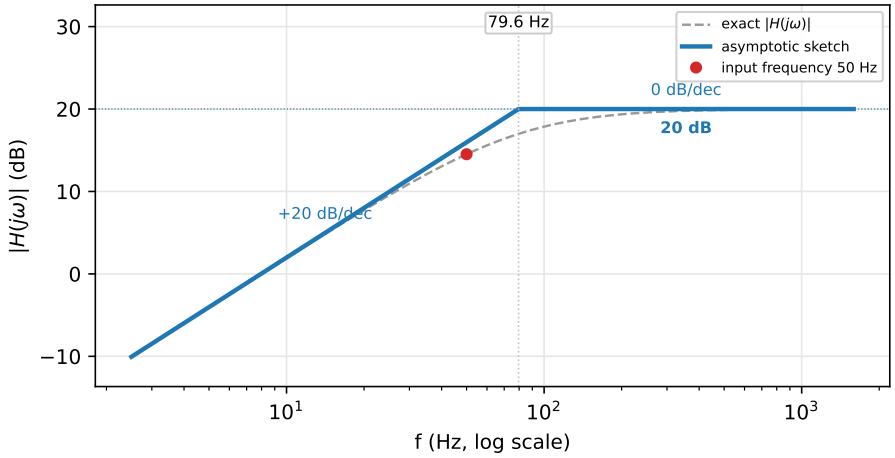
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	500	79.6 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.02\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 79.6$ Hz	+20 dB/dec	20 dB at end
$f > 79.6$ Hz	0 dB/dec (flat)	20 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 10 = 20$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi \cdot 50\,t)$, so $\omega_{in} = 2\pi \cdot 50 = 314$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-10	10	-3.14
$\dfrac{s}{s + 2\pi \cdot 79.6}$	0.532	1.01
total	**5.32** (product)	$\mathbf{-2.13}$ (sum)

$$|H(j\omega_{in})| = 5.32, \qquad \angle H(j\omega_{in}) = -2.13 \text{ rad}$$
$$v_{o,ac}(t) = 5.32 \times 0.2 \sin(2\pi \cdot 50\,t - 2.13) = 1.06 \sin(2\pi \cdot 50\,t - 2.13)$$

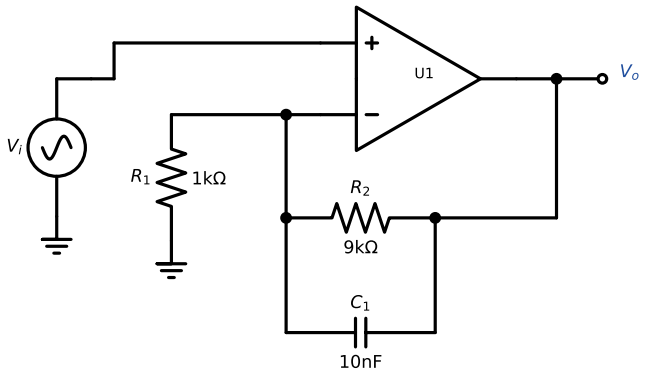
$$\boxed{V_o(t) = 1.06 \sin(2\pi \cdot 50\,t - 2.13) \text{ V}}$$

Answer

$H(s) = -10\,\dfrac{s}{s + 2\pi \cdot 79.6}$ (corners: 79.6 Hz)

$V_o(t) = 1.06 \sin(2\pi \cdot 50\,t - 2.13)$ V

Solution 2A.3 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

The op-amp has negative feedback (its output is connected back to its − input), so with an ideal op-amp (tool T1): (a) no current flows into the + or − input; (b) $V_- = V_+$ (virtual short). The circuit is a standard block, so use its block formula (tool T2).

Step 4 — Transfer function, block by block

Block 1 (U1). Non-inverting block: The + input is the block input: $V_+ = V_i$. The op-amp forces $V_- = V_+$, and R_1, Z_{f1} divide the output: $V_- = \frac{R_1}{R_1+Z_{f1}} V_o$, hence $V_o = \left(1 + \frac{Z_{f1}}{R_1}\right) V_+$.

$$Z_{f1} = R_2 \parallel \frac{1}{sC_1} = \frac{10^8}{s + 11100}$$

$$\frac{V_o}{V_i} = \left(1 + \frac{Z_{f1}}{R_1}\right) = \frac{s + 1.11 \times 10^5}{s + 11100}$$

So:

$$H(s) = \frac{s + 1.11 \times 10^5}{s + 11100}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $s + 1.11 \times 10^5$; $s + 1.11 \times 10^5 = 0 \Rightarrow s = -1.11 \times 10^5$.
- Denominator: $s + 11100$; $s + 11100 = 0 \Rightarrow s = -11100$.

Factored form (leading coefficient in front):

$$H(s) = 1\,\frac{(s + 1.11 \times 10^5)}{(s + 11100)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

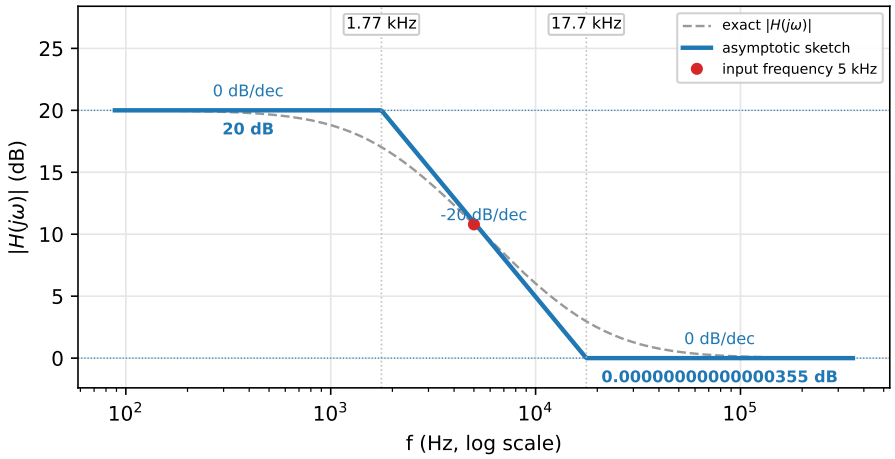
$$H(s) = 10\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1.77\text{k}}}\,\left(1+\frac{s}{2\pi\cdot 17.7\text{k}}\right), \qquad K = 10$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	11100	1.77 kHz	-20 dB/dec
zero	1.11×10^5	17.7 kHz	$+20$ dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 10 = 20\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.77$ kHz	0 dB/dec (flat)	20 dB at end
1.77 kHz – 17.7 kHz	-20 dB/dec	20 dB at start, 0 dB at end
$f > 17.7$ kHz	0 dB/dec (flat)	0 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi\cdot 5000\,t)$, so $\omega_{in} = 2\pi\cdot 5000 = 31400\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
10	10	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.77\text{k}}}$	0.3334	-1.23
$\left(1+\dfrac{s}{2\pi\cdot 17.7\text{k}}\right)$	1.039	0.276
total	**3.465** (product)	$-$**0.955** (sum)

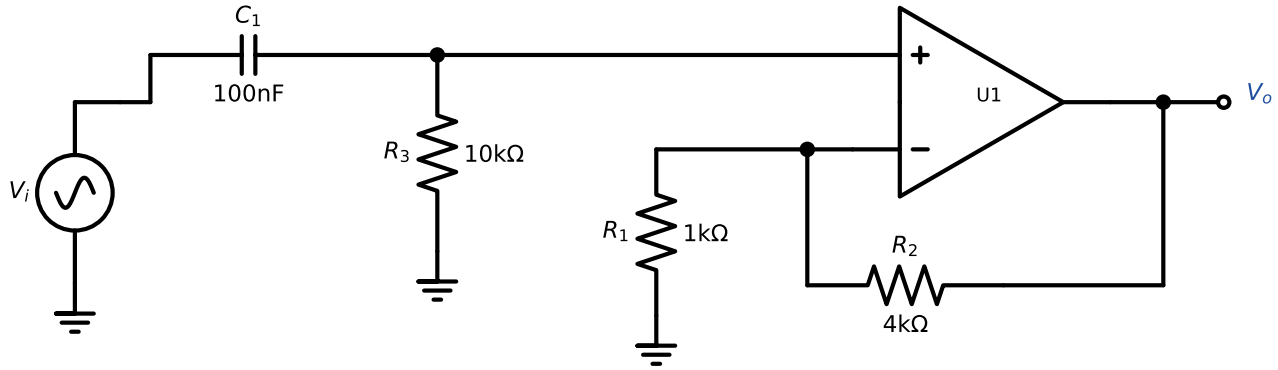
$$|H(j\omega_{in})| = 3.47, \qquad \angle H(j\omega_{in}) = -0.955 \text{ rad}$$
$$v_{o,ac}(t) = 3.47 \times 0.1 \sin(2\pi\cdot 5000\,t - 0.955) = 0.347 \sin(2\pi\cdot 5000\,t - 0.955)$$

$$\boxed{V_o(t) = 0.347 \sin(2\pi\cdot 5000\,t - 0.955) \text{ V}}$$

Answer

$$H(s) = 10\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1.77\text{k}}}\left(1+\frac{s}{2\pi\cdot 17.7\text{k}}\right) \quad \text{(corners: 1.77 kHz, 17.7 kHz)}$$
$$V_o(t) = 0.347 \sin(2\pi\cdot 5000\,t - 0.955) \text{ V}$$

Solution 2A.4 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

The op-amp has negative feedback (its output is connected back to its − input), so with an ideal op-amp (tool T1): (a) no current flows into the + or − input; (b) $V_- = V_+$ (virtual short). The circuit is a standard block, so use its block formula (tool T2).

Step 4 — Transfer function, block by block

Block 1 (U1). Non-inverting block: No current flows into the + input, so R_3 and $\frac{1}{sC_1}$ form a plain voltage divider: $V_+ = \frac{R_3}{\frac{1}{sC_1}+R_3} V_i$. The op-amp forces $V_- = V_+$, and R_1, R_2 divide the output: $V_- = \frac{R_1}{R_1+R_2} V_o$, hence $V_o = \left(1 + \frac{R_2}{R_1}\right) V_+$.

$$\frac{V_o}{V_i} = \frac{R_3}{\frac{1}{sC_1} + R_3}\left(1 + \frac{R_2}{R_1}\right) = \frac{5\,s}{s + 1000}$$

So:

$$H(s) = \frac{5\,s}{s + 1000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow$ 1 zero at $s = 0$.
- Denominator: $s + 1000$; $s + 1000 = 0 \Rightarrow s = -1000$.

Factored form (leading coefficient in front):

$$H(s) = 5\,\frac{s}{(s + 1000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 5\,\frac{s}{s + 2\pi \cdot 159}, \qquad K = 5$$

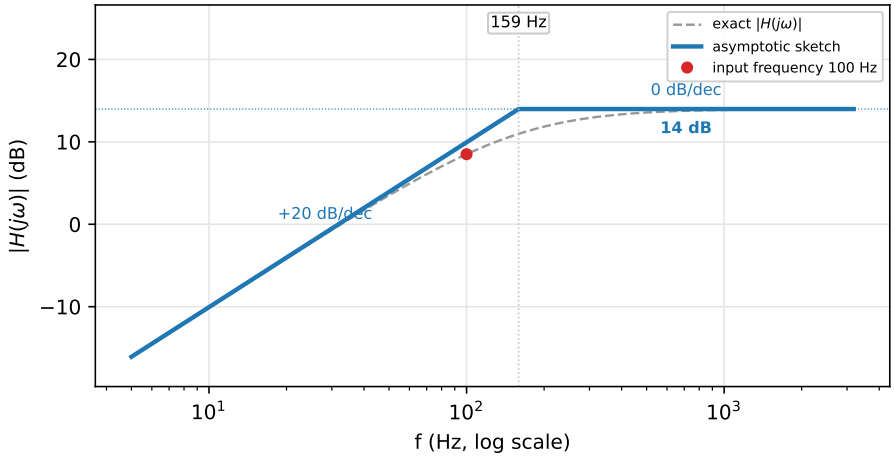
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1000	159 Hz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.005\,s$, so the plot starts with a slope of $+20\,\text{dB/dec}$. Then, moving right, each pole bends the line down by $20\,\text{dB/dec}$ and each zero bends it up by $20\,\text{dB/dec}$ (tool T6):

frequency range	slope	asymptote level
$f < 159$ Hz	$+20$ dB/dec	14 dB at end
$f > 159$ Hz	0 dB/dec (flat)	14 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 5 = 14\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi \cdot 100\,t)$, so $\omega_{in} = 2\pi \cdot 100 = 628\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
5	5	0
$\dfrac{s}{s+2\pi\cdot 159}$	0.532	1.01
total	**2.66** (product)	**1.01** (sum)

$$|H(j\omega_{in})| = 2.66, \qquad \angle H(j\omega_{in}) = 1.01 \text{ rad}$$
$$v_{o,ac}(t) = 2.66 \times 0.2 \sin(2\pi\cdot 100\,t + 1.01) = 0.532 \sin(2\pi\cdot 100\,t + 1.01)$$

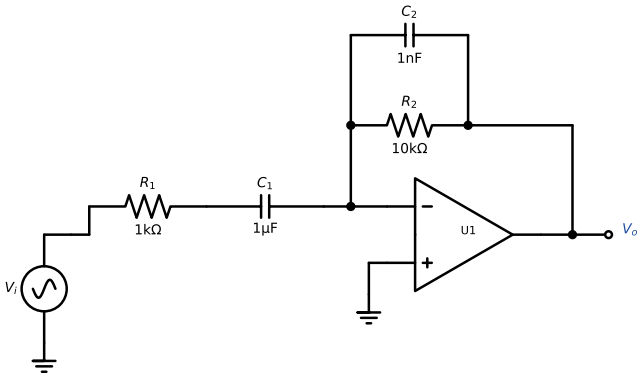
$$\boxed{V_o(t) = 0.532 \sin(2\pi\cdot 100\,t + 1.01) \text{ V}}$$

Answer

$H(s) = 5\,\dfrac{s}{s+2\pi\cdot 159}$ (corners: 159 Hz)

$V_o(t) = 0.532 \sin(2\pi\cdot 100\,t + 1.01)$ V

Solution 2A.5 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

The op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a standard block, so use its block formula (tool T2).

Step 4 — Transfer function, block by block

Block 1 (U1). Inverting block: the $+$ input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{Z_{in1}}$ through Z_{in1} cannot enter the op-amp, so all of it flows through Z_{f1}: $V_o = 0 - Z_{f1} \cdot \frac{V_i}{Z_{in1}}$.

$$Z_{in1} = R_1 + \frac{1}{sC_1} = \frac{1000\,s + 10^6}{s}$$

$$Z_{f1} = R_2 \parallel \frac{1}{sC_2} = \frac{10^9}{s + 10^5}$$

$$\frac{V_o}{V_i} = -\frac{Z_{f1}}{Z_{in1}} = \frac{-10^6\,s}{s^2 + 1.01 \times 10^5\,s + 10^8}$$

So:

$$H(s) = \frac{-10^6\,s}{s^2 + 1.01 \times 10^5\,s + 10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 1.01 \times 10^5\,s + 10^8$; quadratic formula:

$$s = \frac{-1.01 \times 10^5 \pm \sqrt{(1.01 \times 10^5)^2 - 4(10^8)}}{2} = \frac{-1.01 \times 10^5 \pm 99000}{2} \Rightarrow s = -1000,\ -10^5$$

Factored form (leading coefficient in front):

$$H(s) = -10^6 \, \frac{s}{(s+1000)(s+10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -10 \, \frac{s}{s + 2\pi \cdot 159} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 15.9\text{k}}}, \qquad K = -10$$

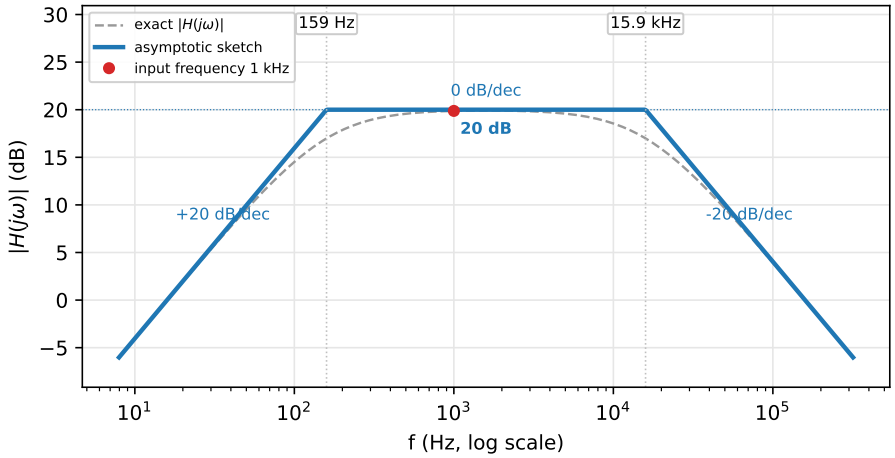
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1000	159 Hz	-20 dB/dec
pole	10^5	15.9 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.01\,s$, so the plot starts with a slope of $+20$ dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 159$ Hz	$+20$ dB/dec	20 dB at end
159 Hz – 15.9 kHz	0 dB/dec (flat)	20 dB at start, 20 dB at end
$f > 15.9$ kHz	-20 dB/dec	20 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 10 = 20$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin(2\pi\cdot 1000\,t)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-10	10	-3.14
$\dfrac{s}{s+2\pi\cdot 159}$	0.9876	0.158
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 15.9\text{k}}}$	0.998	-0.0627
total	**9.856** (product)	$-$**3.05** (sum)

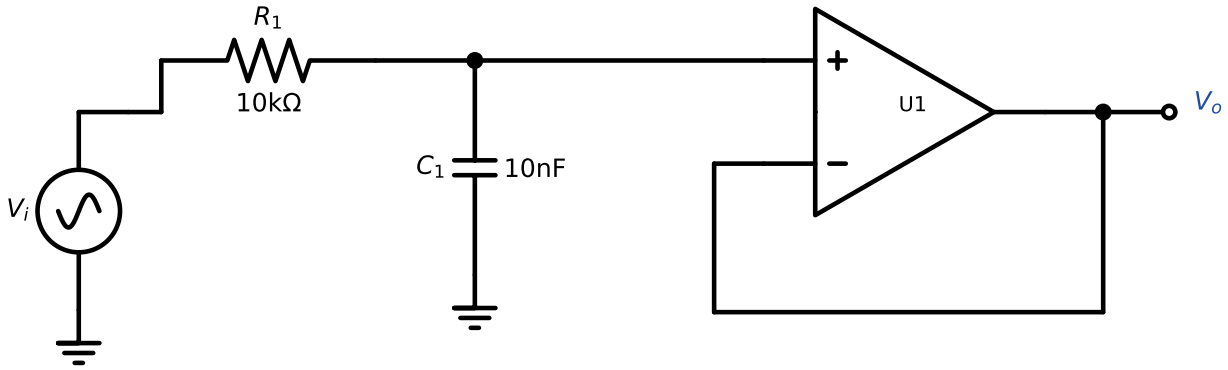
$$|H(j\omega_{in})| = 9.86, \qquad \angle H(j\omega_{in}) = -3.05 \text{ rad}$$
$$v_{o,ac}(t) = 9.86\times 0.05\sin(2\pi\cdot 1000\,t - 3.05) = 0.493\sin(2\pi\cdot 1000\,t - 3.05)$$

$$\boxed{V_o(t) = 0.493\sin(2\pi\cdot 1000\,t - 3.05)\ \text{V}}$$

Answer

$$H(s) = -10\,\frac{s}{s+2\pi\cdot 159}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 15.9\text{k}}} \quad \text{(corners: 159 Hz, 15.9 kHz)}$$
$$V_o(t) = 0.493\sin(2\pi\cdot 1000\,t - 3.05)\ \text{V}$$

Solution 2A.6 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

The op-amp has negative feedback (its output is connected back to its − input), so with an ideal op-amp (tool T1): (a) no current flows into the + or − input; (b) $V_- = V_+$ (virtual short). The circuit is a standard block, so use its block formula (tool T2).

Step 4 — Transfer function, block by block

Block 1 (U1). Non-inverting block: No current flows into the + input, so $\frac{1}{sC_1}$ and R_1 form a plain voltage divider: $V_+ = \frac{\frac{1}{sC_1}}{R_1+\frac{1}{sC_1}} V_i$. The output is tied back to the − input, so $V_o = V_- = V_+$ (voltage follower).

$$\frac{V_o}{V_i} = \frac{\frac{1}{sC_1}}{R_1 + \frac{1}{sC_1}} = \frac{10000}{s + 10000}$$

So:

$$H(s) = \frac{10000}{s + 10000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 10000$; $s + 10000 = 0 \Rightarrow s = -10000$.

Factored form (leading coefficient in front):

$$H(s) = 10000\,\frac{1}{(s + 10000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 1\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}}, \qquad K = 1$$

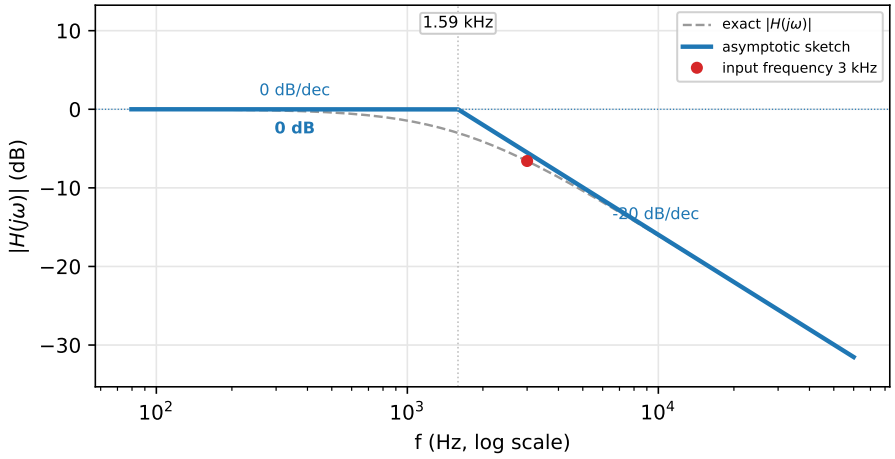
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	10000	1.59 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = 0\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.59$ kHz	0 dB/dec (flat)	0 dB at end
$f > 1.59$ kHz	-20 dB/dec	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 1 = 0\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 1\cos(2\pi \cdot 3000\,t)$, so $\omega_{in} = 2\pi \cdot 3000 = 18800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 1.59\text{k}}}$	0.4686	-1.08
total	**0.4686** (product)	$\mathbf{-1.08}$ (sum)

$$|H(j\omega_{in})| = 0.469, \qquad \angle H(j\omega_{in}) = -1.08 \text{ rad}$$

$$v_{o,ac}(t) = 0.469 \times 1 \cos\left(2\pi \cdot 3000\, t - 1.08\right) = 0.469 \cos\left(2\pi \cdot 3000\, t - 1.08\right)$$

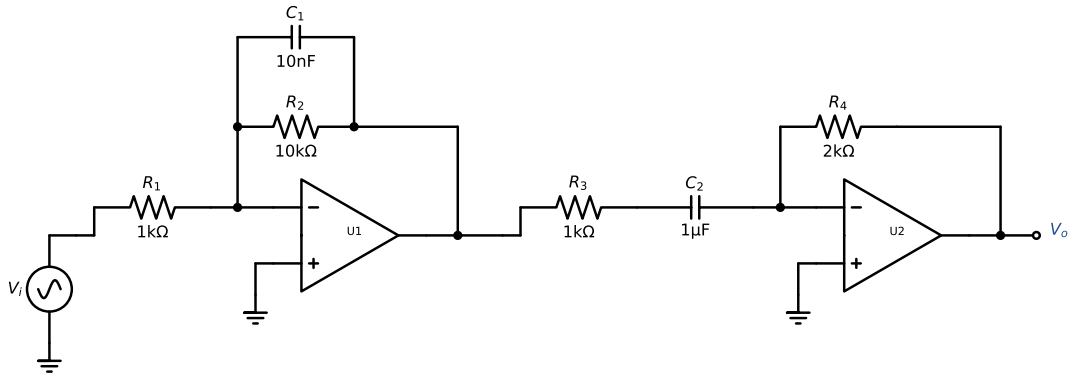
$$\boxed{V_o(t) = 0.469 \cos\left(2\pi \cdot 3000\, t - 1.08\right) \text{ V}}$$

Answer

$$H(s) = 1 \frac{1}{1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}} \quad \text{(corners: 1.59 kHz)}$$

$$V_o(t) = 0.469 \cos\left(2\pi \cdot 3000\, t - 1.08\right) \text{ V}$$

Solution 2A.7 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a cascade of standard blocks, so each block is solved with its block formula (tool T2).

Step 4 — Transfer function, block by block

Each op-amp output is an ideal voltage source, so a block's output does not change when the next block is connected: the blocks can be solved one at a time and multiplied.

$$H(s) = \frac{V_o}{V_i} = \frac{V_{o1}}{V_i} \cdot \frac{V_o}{V_{o1}}$$

Block 1 (U1). Inverting block: the $+$ input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{R_1}$ through R_1 cannot enter the op-amp, so all of it flows through Z_{f1}: $V_{o1} = 0 - Z_{f1} \cdot \frac{V_i}{R_1}$.

$$Z_{f1} = R_2 \parallel \frac{1}{sC_1} = \frac{10^8}{s + 10000}$$

$$\frac{V_{o1}}{V_i} = -\frac{Z_{f1}}{R_1} = \frac{-10^5}{s + 10000}$$

Block 2 (U2). Inverting block: the $+$ input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1} - 0}{Z_{in2}}$ through Z_{in2} cannot enter the op-amp, so all of it flows through R_4: $V_o = 0 - R_4 \cdot \frac{V_{o1}}{Z_{in2}}$.

$$Z_{in2} = R_3 + \frac{1}{sC_2} = \frac{1000\,s + 10^6}{s}$$

$$\frac{V_o}{V_{o1}} = -\frac{R_4}{Z_{in2}} = \frac{-2\,s}{s + 1000}$$

Multiply the blocks and simplify:

$$H(s) = \frac{2 \times 10^5\, s}{s^2 + 11000\, s + 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 11000\,s + 10^7$; quadratic formula:

$$s = \frac{-11000 \pm \sqrt{(11000)^2 - 4(10^7)}}{2} = \frac{-11000 \pm 9000}{2} \Rightarrow s = -1000,\ -10000$$

Factored form (leading coefficient in front):

$$H(s) = 2 \times 10^5 \, \frac{s}{(s + 1000)(s + 10000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 20 \, \frac{s}{s + 2\pi \cdot 159} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}}, \qquad K = 20$$

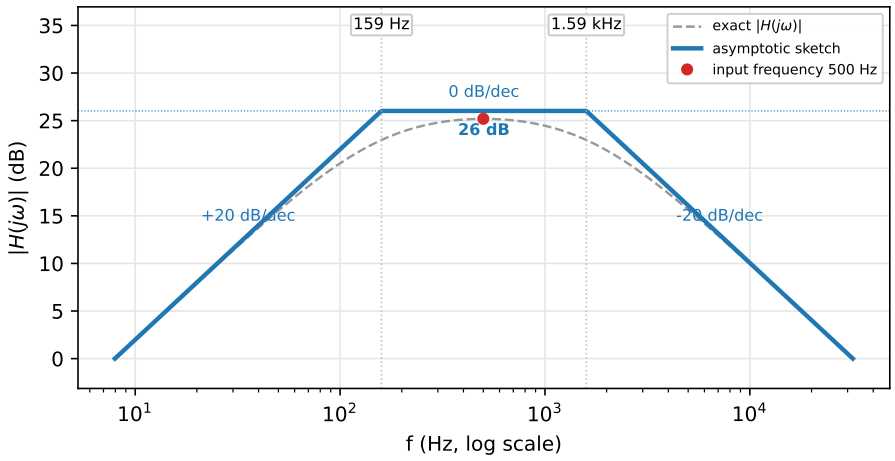
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1000	159 Hz	−20 dB/dec
pole	10000	1.59 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.02\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 159 Hz	+20 dB/dec	26 dB at end
159 Hz – 1.59 kHz	0 dB/dec (flat)	26 dB at start, 26 dB at end
$f >$ 1.59 kHz	−20 dB/dec	26 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 20 = 26$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.02\sin(2\pi \cdot 500\,t)$, so $\omega_{in} = 2\pi \cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
20	20	0
$\dfrac{s}{s+2\pi\cdot 159}$	0.9529	0.308
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}$	0.954	−0.304
total	**18.18** (product)	**0.00377** (sum)

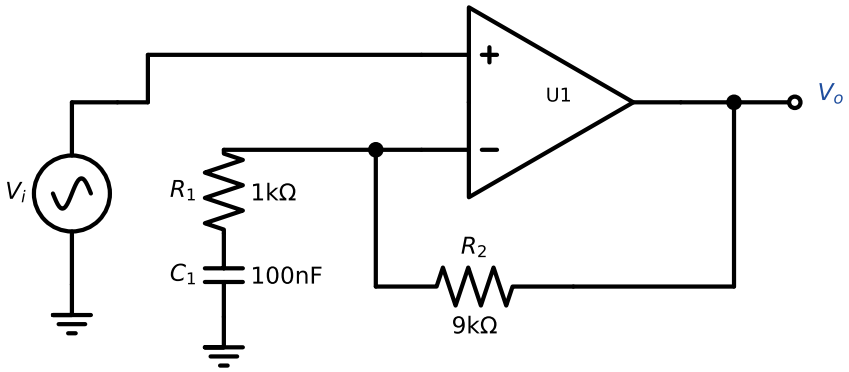
$$|H(j\omega_{in})| = 18.2, \qquad \angle H(j\omega_{in}) = 0.00377 \text{ rad}$$
$$v_{o,ac}(t) = 18.2 \times 0.02 \sin(2\pi \cdot 500\,t + 0.00377) = 0.364 \sin(2\pi \cdot 500\,t + 0.00377)$$

$$\boxed{V_o(t) = 0.364 \sin(2\pi \cdot 500\,t + 0.00377)\ \text{V}}$$

Answer

$$H(s) = 20\,\frac{s}{s+2\pi\cdot 159}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}} \quad \text{(corners: 159 Hz, 1.59 kHz)}$$
$$V_o(t) = 0.364 \sin(2\pi \cdot 500\,t + 0.00377)\ \text{V}$$

Solution 2A.8 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

The op-amp has negative feedback (its output is connected back to its − input), so with an ideal op-amp (tool T1): (a) no current flows into the + or − input; (b) $V_- = V_+$ (virtual short). The circuit is a standard block, so use its block formula (tool T2).

Step 4 — Transfer function, block by block

Block 1 (U1). Non-inverting block: The + input is the block input: $V_+ = V_i$. The op-amp forces $V_- = V_+$, and Z_{g1}, R_2 divide the output: $V_- = \frac{Z_{g1}}{Z_{g1}+R_2} V_o$, hence $V_o = \left(1 + \frac{R_2}{Z_{g1}}\right) V_+$.

$$Z_{g1} = R_1 + \frac{1}{sC_1} = \frac{1000\,s + 10^7}{s}$$

$$\frac{V_o}{V_i} = \left(1 + \frac{R_2}{Z_{g1}}\right) = \frac{10\,s + 10000}{s + 10000}$$

So:

$$H(s) = \frac{10\,s + 10000}{s + 10000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $10\,s + 10000$; divide by the leading coefficient 10 (this does not move the roots): $s + 1000$; $s + 1000 = 0 \Rightarrow s = -1000$.
- Denominator: $s + 10000$; $s + 10000 = 0 \Rightarrow s = -10000$.

Factored form (leading coefficient in front):

$$H(s) = 10\,\frac{(s + 1000)}{(s + 10000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

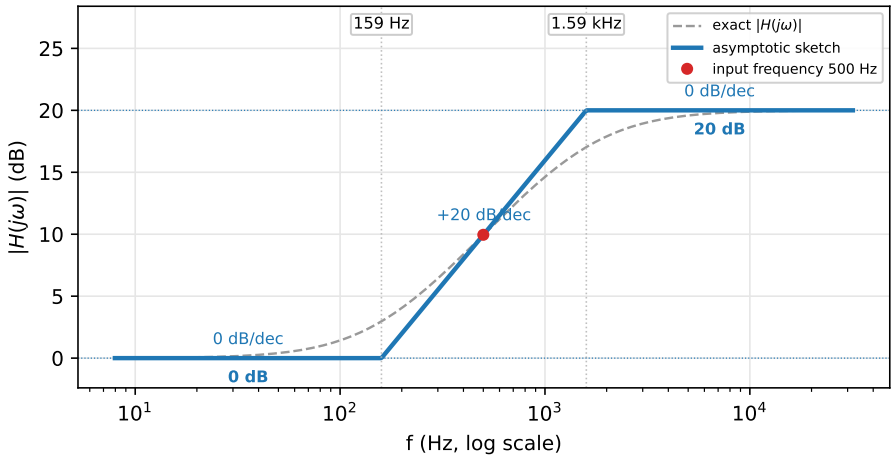
$$H(s) = 1\left(1+\frac{s}{2\pi\cdot 159}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}, \qquad K = 1$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
zero	1000	159 Hz	+20 dB/dec
pole	10000	1.59 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = 0\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 159$ Hz	0 dB/dec (flat)	0 dB at end
159 Hz – 1.59 kHz	+20 dB/dec	0 dB at start, 20 dB at end
$f > 1.59$ kHz	0 dB/dec (flat)	20 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi\cdot 500\,t)$, so $\omega_{in} = 2\pi\cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\left(1+\dfrac{s}{2\pi\cdot 159}\right)$	3.297	1.26
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}$	0.954	-0.304
total	**3.145** (product)	**0.958** (sum)

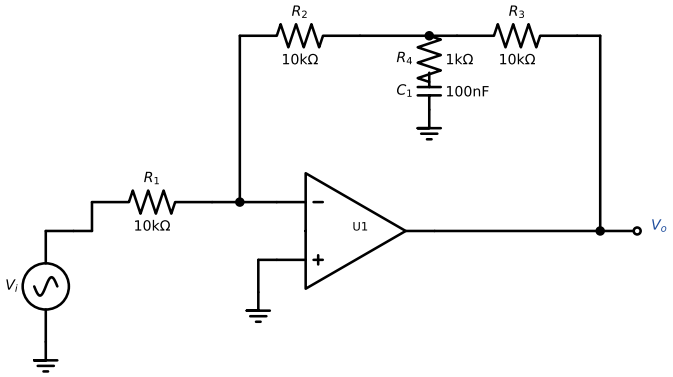
$$|H(j\omega_{in})| = 3.15, \qquad \angle H(j\omega_{in}) = 0.958 \text{ rad}$$
$$v_{o,ac}(t) = 3.15 \times 0.1 \sin(2\pi\cdot 500\,t + 0.958) = 0.315 \sin(2\pi\cdot 500\,t + 0.958)$$

$$\boxed{V_o(t) = 0.315 \sin(2\pi\cdot 500\,t + 0.958) \text{ V}}$$

Answer

$$H(s) = 1\left(1+\frac{s}{2\pi\cdot 159}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}} \quad \text{(corners: 159 Hz, 1.59 kHz)}$$
$$V_o(t) = 0.315 \sin(2\pi\cdot 500\,t + 0.958) \text{ V}$$

Solution 2A.9 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

The op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a standard block, so use its block formula (tool T2).

Step 4 — Transfer function, block by block

Block 1 (U1). T-network block. Virtual ground: $V_- = 0$, so $I = \frac{V_i}{R_1}$ flows through R_1 and (no current into the op-amp) on through R_2. The T node is at $V_T = 0 - R_2 I$. Z_{c1} takes $\frac{V_T}{Z_{c1}}$ to ground, so KCL at the T node leaves $I_b = I - \frac{V_T}{Z_{c1}} = I\left(1 + \frac{R_2}{Z_{c1}}\right)$ for R_3. Then $V_o = V_T - R_3 I_b = -\left(R_2 + R_3 + \frac{R_2 R_3}{Z_{c1}}\right) I$.

$$Z_{c1} = R_4 + \frac{1}{sC_1} = \frac{1000\,s + 10^7}{s}$$

$$\frac{V_o}{V_i} = -\frac{R_2 + R_3 + \frac{R_2 R_3}{Z_{c1}}}{R_1} = \frac{-12\,s - 20000}{s + 10000}$$

So:

$$H(s) = \frac{-12\,s - 20000}{s + 10000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $-12\,s - 20000$; divide by the leading coefficient -12 (this does not move the roots): $s + 1670$; $s + 1670 = 0 \Rightarrow s = -1670$.
- Denominator: $s + 10000$; $s + 10000 = 0 \Rightarrow s = -10000$.

Factored form (leading coefficient in front):

$$H(s) = -12\,\frac{(s+1670)}{(s+10000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

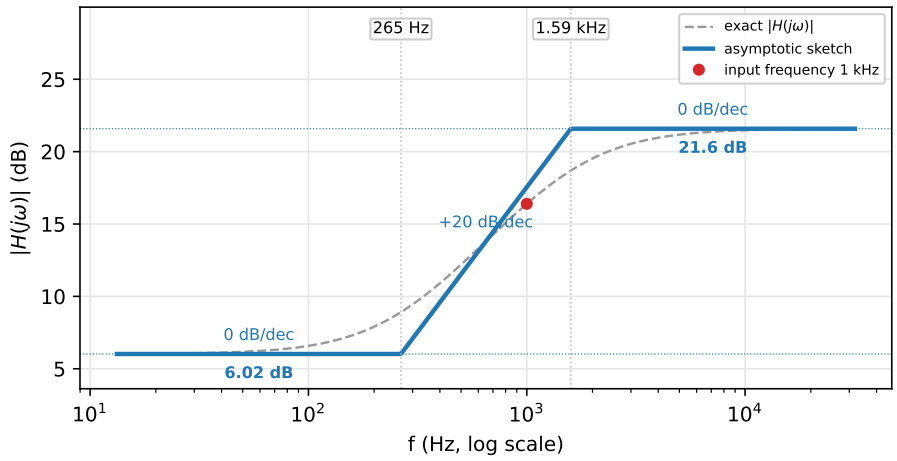
$$H(s) = -2\left(1+\frac{s}{2\pi\cdot 265}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}, \qquad K=-2$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
zero	1670	265 Hz	+20 dB/dec
pole	10000	1.59 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 2 = 6.02\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 265$ Hz	0 dB/dec (flat)	6.02 dB at end
265 Hz – 1.59 kHz	+20 dB/dec	6.02 dB at start, 21.6 dB at end
$f > 1.59$ kHz	0 dB/dec (flat)	21.6 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi\cdot 1000\,t)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-2	2	-3.14
$\left(1+\dfrac{s}{2\pi\cdot 265}\right)$	3.9	1.31
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}$	0.8467	-0.561
total	**6.605** (product)	$\mathbf{-2.39}$ (sum)

$$|H(j\omega_{in})| = 6.61, \qquad \angle H(j\omega_{in}) = -2.39 \text{ rad}$$
$$v_{o,ac}(t) = 6.61 \times 0.1 \sin(2\pi\cdot 1000\,t - 2.39) = 0.661 \sin(2\pi\cdot 1000\,t - 2.39)$$

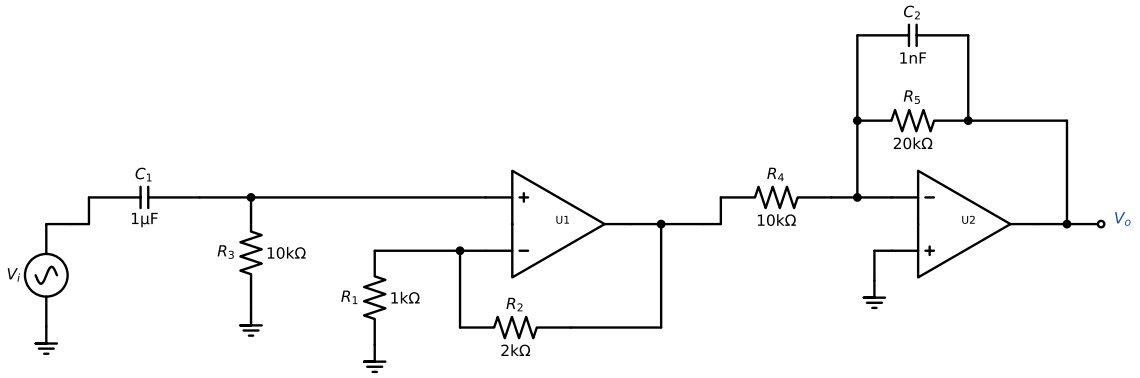
$$\boxed{V_o(t) = 0.661 \sin(2\pi\cdot 1000\,t - 2.39) \text{ V}}$$

Answer

$H(s) = -2\left(1+\dfrac{s}{2\pi\cdot 265}\right)\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}$ (corners: 265 Hz, 1.59 kHz)

$V_o(t) = 0.661 \sin(2\pi\cdot 1000\,t - 2.39)$ V

Solution 2A.10 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a cascade of standard blocks, so each block is solved with its block formula (tool T2).

Step 4 — Transfer function, block by block

Each op-amp output is an ideal voltage source, so a block's output does not change when the next block is connected: the blocks can be solved one at a time and multiplied.

$$H(s) = \frac{V_o}{V_i} = \frac{V_{o1}}{V_i} \cdot \frac{V_o}{V_{o1}}$$

Block 1 (U1). Non-inverting block: No current flows into the $+$ input, so R_3 and $\frac{1}{sC_1}$ form a plain voltage divider: $V_+ = \frac{R_3}{\frac{1}{sC_1}+R_3} V_i$. The op-amp forces $V_- = V_+$, and R_1, R_2 divide the output: $V_- = \frac{R_1}{R_1+R_2} V_{o1}$, hence $V_{o1} = \left(1 + \frac{R_2}{R_1}\right) V_+$.

$$\frac{V_{o1}}{V_i} = \frac{R_3}{\frac{1}{sC_1} + R_3} \left(1 + \frac{R_2}{R_1}\right) = \frac{3\,s}{s + 100}$$

Block 2 (U2). Inverting block: the $+$ input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1}-0}{R_4}$ through R_4 cannot enter the op-amp, so all of it flows through Z_{f2}: $V_o = 0 - Z_{f2} \cdot \frac{V_{o1}}{R_4}$.

$$Z_{f2} = R_5 \parallel \frac{1}{sC_2} = \frac{10^9}{s + 50000}$$

$$\frac{V_o}{V_{o1}} = -\frac{Z_{f2}}{R_4} = \frac{-10^5}{s + 50000}$$

Multiply the blocks and simplify:

$$H(s) = \frac{-3 \times 10^5\, s}{s^2 + 50100\, s + 5 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 50100\,s + 5 \times 10^6$; quadratic formula:

$$s = \frac{-50100 \pm \sqrt{(50100)^2 - 4(5 \times 10^6)}}{2} = \frac{-50100 \pm 49900}{2} \Rightarrow s = -100,\ -50000$$

Factored form (leading coefficient in front):

$$H(s) = -3 \times 10^5 \, \frac{s}{(s + 100)(s + 50000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -6 \, \frac{s}{s + 2\pi \cdot 15.9} \, \frac{1}{1 + \dfrac{s}{2\pi \cdot 7.96\text{k}}}, \qquad K = -6$$

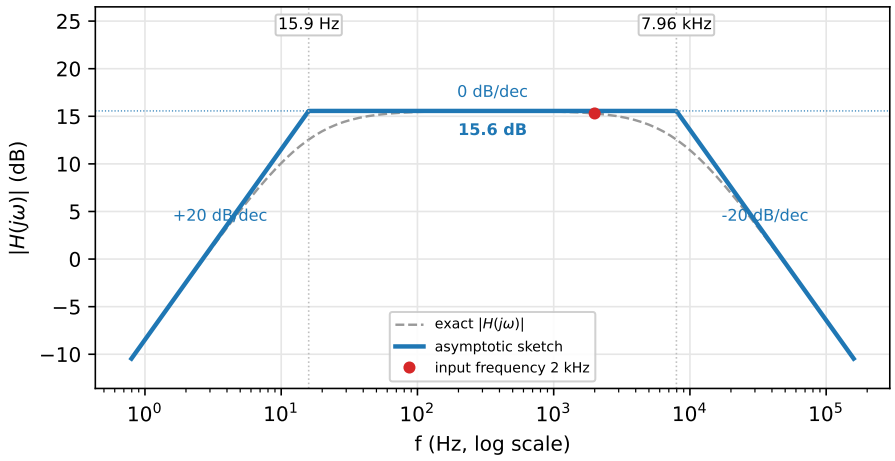
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	100	15.9 Hz	−20 dB/dec
pole	50000	7.96 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.06\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 15.9 Hz	+20 dB/dec	15.6 dB at end
15.9 Hz – 7.96 kHz	0 dB/dec (flat)	15.6 dB at start, 15.6 dB at end
$f >$ 7.96 kHz	−20 dB/dec	15.6 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 6 = 15.6$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-6	6	-3.14
$\dfrac{s}{s+2\pi\cdot 15.9}$	1	0.00796
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 7.96\text{k}}}$	0.9698	-0.246
total	**5.819** (product)	$\mathbf{-3.38}$ (sum)

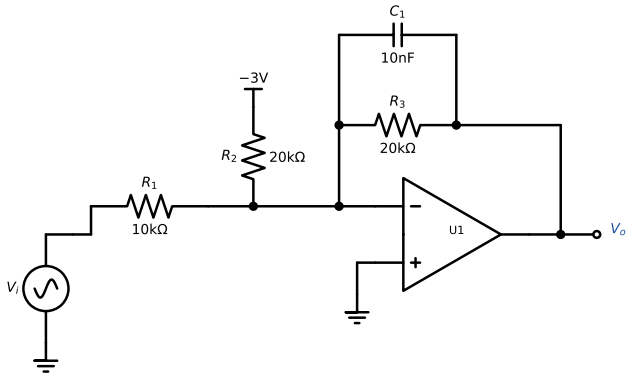
$$|H(j\omega_{in})| = 5.82, \qquad \angle H(j\omega_{in}) = -3.38 \text{ rad}$$
$$v_{o,ac}(t) = 5.82 \times 0.1 \sin(2\pi \cdot 2000\,t - 3.38) = 0.582 \sin(2\pi \cdot 2000\,t - 3.38)$$

$$\boxed{V_o(t) = 0.582 \sin(2\pi \cdot 2000\,t - 3.38)\ \text{V}}$$

Answer

$$H(s) = -6\,\frac{s}{s+2\pi\cdot 15.9}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 7.96\text{k}}} \quad \text{(corners: 15.9 Hz, 7.96 kHz)}$$
$$V_o(t) = 0.582 \sin(2\pi \cdot 2000\,t - 3.38)\ \text{V}$$

Solution 2A.11 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supply $-3\,\text{V}$. Both DC and AC are present, so use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. The DC part is Step 2, the AC part Steps 3–7.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: $V_i = 0$ and every capacitor is an open circuit (no inductors appear in these circuits). A block passes DC with its gain at $s = 0$, i.e. $H_k(0)$.

1. Block 1 (summer): its input has no DC. Only the supply drives a DC current, $\frac{V_{DC}}{R_2}$, through Z_{f1} (at DC $R_2 = 20\,\text{k}\Omega$, $Z_{f1} = 20\,\text{k}\Omega$):

$$V_o\big|_{dc} = -Z_{f1}\frac{V_{DC}}{R_2} = -20\,\text{k}\cdot\frac{-3}{20\,\text{k}} = 3\,\text{V}$$

$$\boxed{V_{o,dc} = 3\,\text{V}}$$

Step 3 — Op-amp rules

The op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a standard block, so use its block formula (tool T2).

Step 4 — Transfer function, block by block

Block 1 (U1). Summing block: $V_- = V_+ = 0$ (virtual ground), so the input currents $\frac{V_i}{R_1}$ and $\frac{V_{DC}}{R_2}$ simply add and both flow through Z_{f1}: $V_o = -Z_{f1}\left(\frac{V_i}{R_1} + \frac{V_{DC}}{R_2}\right)$. For the AC part the supply is zero, so R_2 (from virtual ground to AC ground) carries no AC current and drops out.

$$Z_{f1} = R_3 \parallel \frac{1}{sC_1} = \frac{10^8}{s + 5000}$$

$$\frac{V_o}{V_i} = -\frac{Z_{f1}}{R_1} = \frac{-10000}{s + 5000}$$

So:

$$H(s) = \frac{-10000}{s + 5000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 5000$; $s + 5000 = 0 \Rightarrow s = -5000$.

Factored form (leading coefficient in front):

$$H(s) = -10000\,\frac{1}{(s + 5000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -2\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 796}}, \qquad K = -2$$

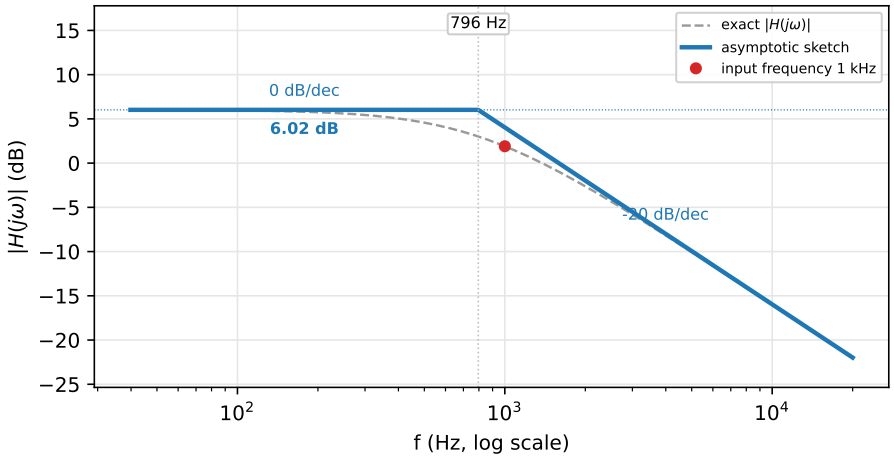
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	5000	796 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 2 = 6.02\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 796$ Hz	0 dB/dec (flat)	6.02 dB at end
$f > 796$ Hz	−20 dB/dec	6.02 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 2 = 6.02\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-2	2	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 796}}$	0.6227	-0.899
total	**1.245** (product)	$-$**4.04** (sum)

$$|H(j\omega_{in})| = 1.25, \qquad \angle H(j\omega_{in}) = -4.04 \text{ rad}$$
$$v_{o,ac}(t) = 1.25 \times 0.5\,\sin(2\pi \cdot 1000\,t - 4.04) = 0.623\,\sin(2\pi \cdot 1000\,t - 4.04)$$

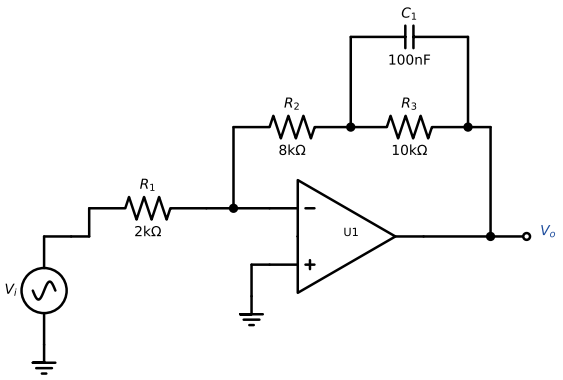
Add the DC part from Step 2 (superposition):

$$\boxed{V_o(t) = 3 + 0.623\,\sin(2\pi \cdot 1000\,t - 4.04)\ \text{V}}$$

Answer

$$H(s) = -2\,\frac{1}{1+\dfrac{s}{2\pi \cdot 796}} \quad \text{(corners: 796 Hz)}$$
$$V_o(t) = 3 + 0.623\,\sin(2\pi \cdot 1000\,t - 4.04)\ \text{V}$$

Solution 2A.12 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

The op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a standard block, so use its block formula (tool T2).

Step 4 — Transfer function, block by block

Block 1 (U1). Inverting block: the $+$ input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{R_1}$ through R_1 cannot enter the op-amp, so all of it flows through Z_{f1}: $V_o = 0 - Z_{f1} \cdot \frac{V_i}{R_1}$.

$$Z_{f1} = R_2 + \left(R_3 \parallel \frac{1}{sC_1} \right) = \frac{8000\, s + 1.8 \times 10^7}{s + 1000}$$

$$\frac{V_o}{V_i} = -\frac{Z_{f1}}{R_1} = \frac{-4\, s - 9000}{s + 1000}$$

So:

$$H(s) = \frac{-4\, s - 9000}{s + 1000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $-4\, s - 9000$; divide by the leading coefficient -4 (this does not move the roots): $s + 2250$; $s + 2250 = 0 \Rightarrow s = -2250$.
- Denominator: $s + 1000$; $s + 1000 = 0 \Rightarrow s = -1000$.

Factored form (leading coefficient in front):

$$H(s) = -4\, \frac{(s + 2250)}{(s + 1000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

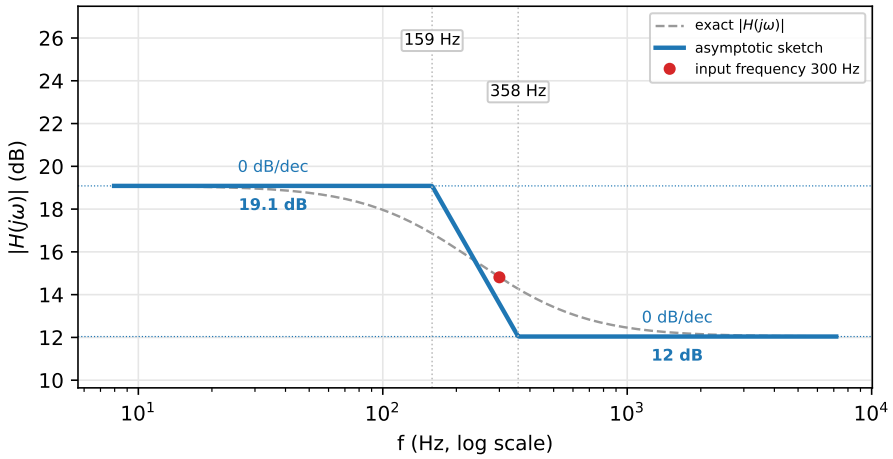
$$H(s) = -9\,\frac{1}{1+\dfrac{s}{2\pi\cdot 159}}\,\left(1+\frac{s}{2\pi\cdot 358}\right), \qquad K = -9$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1000	159 Hz	-20 dB/dec
zero	2250	358 Hz	$+20$ dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 9 = 19.1\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 159$ Hz	0 dB/dec (flat)	19.1 dB at end
159 Hz – 358 Hz	-20 dB/dec	19.1 dB at start, 12 dB at end
$f > 358$ Hz	0 dB/dec (flat)	12 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi\cdot 300\,t)$, so $\omega_{in} = 2\pi\cdot 300 = 1880\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-9	9	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 159}}$	0.4686	-1.08
$\left(1+\dfrac{s}{2\pi\cdot 358}\right)$	1.305	0.697
total	**5.502** (product)	$\mathbf{-3.53}$ (sum)

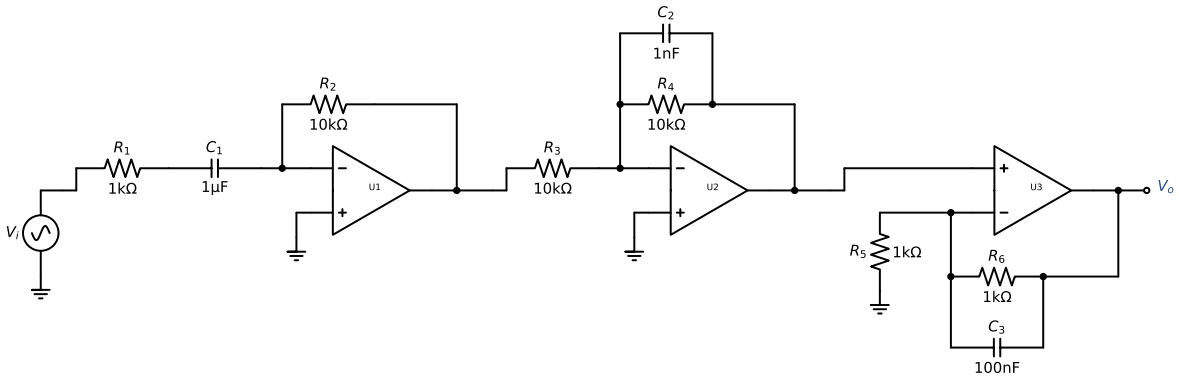
$$|H(j\omega_{in})| = 5.5, \qquad \angle H(j\omega_{in}) = -3.53 \text{ rad}$$
$$v_{o,ac}(t) = 5.5 \times 0.2 \sin(2\pi\cdot 300\,t - 3.53) = 1.1 \sin(2\pi\cdot 300\,t - 3.53)$$

$$\boxed{V_o(t) = 1.1 \sin(2\pi\cdot 300\,t - 3.53) \text{ V}}$$

Answer

$$H(s) = -9\,\frac{1}{1+\dfrac{s}{2\pi\cdot 159}}\left(1+\frac{s}{2\pi\cdot 358}\right) \quad \text{(corners: 159 Hz, 358 Hz)}$$
$$V_o(t) = 1.1 \sin(2\pi\cdot 300\,t - 3.53) \text{ V}$$

Solution 2A.13 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a cascade of standard blocks, so each block is solved with its block formula (tool T2).

Step 4 — Transfer function, block by block

Each op-amp output is an ideal voltage source, so a block's output does not change when the next block is connected: the blocks can be solved one at a time and multiplied.

$$H(s) = \frac{V_o}{V_i} = \frac{V_{o1}}{V_i} \cdot \frac{V_{o2}}{V_{o1}} \cdot \frac{V_o}{V_{o2}}$$

Block 1 (U1). Inverting block: the $+$ input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{Z_{in1}}$ through Z_{in1} cannot enter the op-amp, so all of it flows through R_2: $V_{o1} = 0 - R_2 \cdot \frac{V_i}{Z_{in1}}$.

$$Z_{in1} = R_1 + \frac{1}{sC_1} = \frac{1000\,s + 10^6}{s}$$

$$\frac{V_{o1}}{V_i} = -\frac{R_2}{Z_{in1}} = \frac{-10\,s}{s + 1000}$$

Block 2 (U2). Inverting block: the $+$ input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1} - 0}{R_3}$ through R_3 cannot enter the op-amp, so all of it flows through Z_{f2}: $V_{o2} = 0 - Z_{f2} \cdot \frac{V_{o1}}{R_3}$.

$$Z_{f2} = R_4 \parallel \frac{1}{sC_2} = \frac{10^9}{s + 10^5}$$

$$\frac{V_{o2}}{V_{o1}} = -\frac{Z_{f2}}{R_3} = \frac{-10^5}{s + 10^5}$$

Block 3 (U3). Non-inverting block: The $+$ input is the block input: $V_+ = V_{o2}$. The op-amp forces $V_- = V_+$, and R_5, Z_{f3} divide the output: $V_- = \frac{R_5}{R_5 + Z_{f3}} V_o$, hence $V_o = \left(1 + \frac{Z_{f3}}{R_5}\right) V_+$.

$$Z_{f3} = R_6 \parallel \frac{1}{sC_3} = \frac{10^7}{s + 10000}$$

$$\frac{V_o}{V_{o2}} = \left(1 + \frac{Z_{f3}}{R_5}\right) = \frac{s + 20000}{s + 10000}$$

Multiply the blocks and simplify:

$$H(s) = \frac{10^6\, s^2 + 2 \times 10^{10}\, s}{s^3 + 1.11 \times 10^5\, s^2 + 1.11 \times 10^9\, s + 10^{12}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $10^6\, s + 2 \times 10^{10}$; divide by the leading coefficient 10^6 (this does not move the roots): $s + 20000$; $s + 20000 = 0 \Rightarrow s = -20000$.
- Denominator: $s^3 + 1.11 \times 10^5\, s^2 + 1.11 \times 10^9\, s + 10^{12}$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -1000,\ -10000,\ -100000$.

Factored form (leading coefficient in front):

$$H(s) = 10^6\, \frac{s(s + 20000)}{(s + 1000)(s + 10000)(s + 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

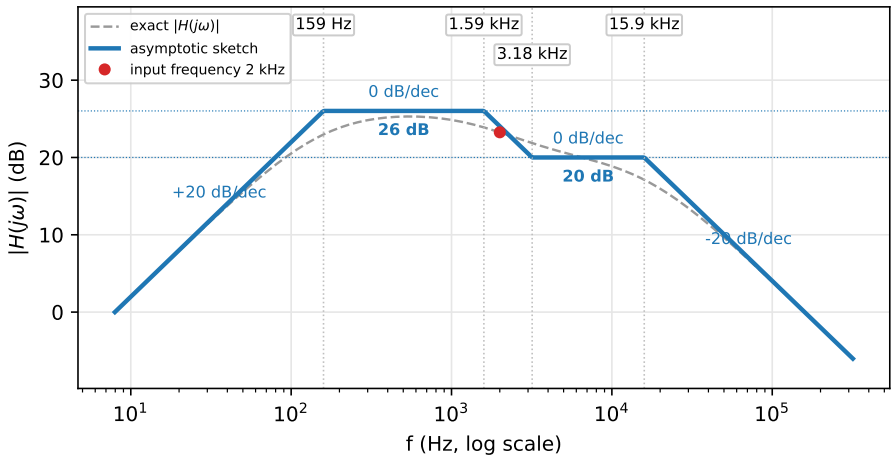
$$H(s) = 20\, \frac{s}{s + 2\pi \cdot 159}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}}\, \left(1 + \frac{s}{2\pi \cdot 3.18\text{k}}\right)\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 15.9\text{k}}}, \qquad K = 20$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1000	159 Hz	−20 dB/dec
pole	10000	1.59 kHz	−20 dB/dec
zero	20000	3.18 kHz	+20 dB/dec
pole	10^5	15.9 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.02\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 159 Hz	+20 dB/dec	26 dB at end
159 Hz – 1.59 kHz	0 dB/dec (flat)	26 dB at start, 26 dB at end
1.59 kHz – 3.18 kHz	−20 dB/dec	26 dB at start, 20 dB at end
3.18 kHz – 15.9 kHz	0 dB/dec (flat)	20 dB at start, 20 dB at end
$f >$ 15.9 kHz	−20 dB/dec	20 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.01\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
20	20	0
$\dfrac{s}{s+2\pi\cdot 159}$	0.9968	0.0794
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}$	0.6227	−0.899
$\left(1+\dfrac{s}{2\pi\cdot 3.18\text{k}}\right)$	1.181	0.561
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 15.9\text{k}}}$	0.9922	−0.125
total	**14.55** (product)	**−0.383** (sum)

$$|H(j\omega_{in})| = 14.5, \qquad \angle H(j\omega_{in}) = -0.383 \text{ rad}$$
$$v_{o,ac}(t) = 14.5 \times 0.01 \sin(2\pi \cdot 2000\,t - 0.383) = 0.145 \sin(2\pi \cdot 2000\,t - 0.383)$$

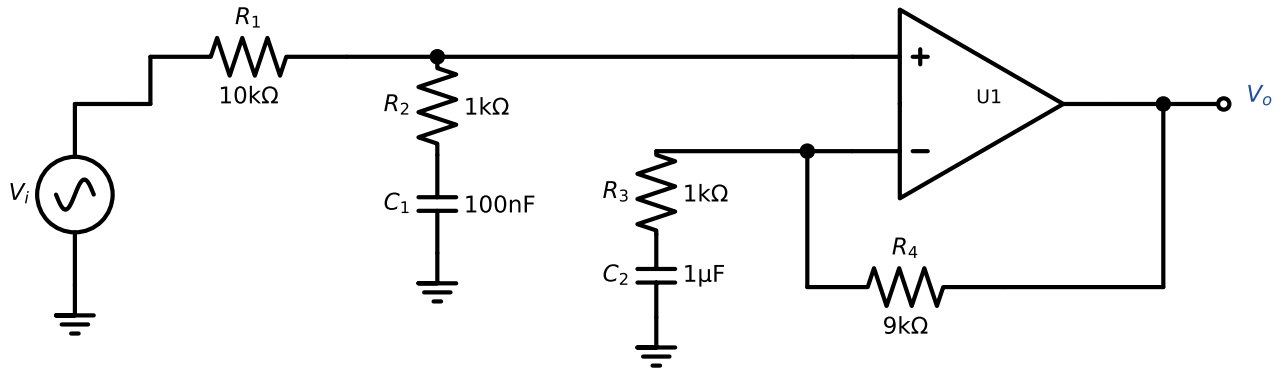
$$\boxed{V_o(t) = 0.145 \sin(2\pi \cdot 2000\,t - 0.383) \text{ V}}$$

Answer

$$H(s) = 20\,\frac{s}{s+2\pi\cdot 159}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}\left(1+\frac{s}{2\pi\cdot 3.18\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 15.9\text{k}}}$$ (corners: 159 Hz, 1.59 kHz, 3.18 kHz, 15.9 kHz)

$V_o(t) = 0.145\,\sin\left(2\pi\cdot 2000\,t - 0.383\right)$ V

Solution 2A.14 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

The op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a standard block, so use its block formula (tool T2).

Step 4 — Transfer function, block by block

Block 1 (U1). Non-inverting block: No current flows into the $+$ input, so Z_{b1} and R_1 form a plain voltage divider: $V_+ = \frac{Z_{b1}}{R_1+Z_{b1}} V_i$. The op-amp forces $V_- = V_+$, and Z_{g1}, R_4 divide the output: $V_- = \frac{Z_{g1}}{Z_{g1}+R_4} V_o$, hence $V_o = \left(1 + \frac{R_4}{Z_{g1}}\right) V_+$.

$$Z_{b1} = R_2 + \frac{1}{sC_1} = \frac{1000\,s + 10^7}{s}$$

$$Z_{g1} = R_3 + \frac{1}{sC_2} = \frac{1000\,s + 10^6}{s}$$

$$\frac{V_o}{V_i} = \frac{Z_{b1}}{R_1 + Z_{b1}} \left(1 + \frac{R_4}{Z_{g1}}\right) = \frac{0.909\,s^2 + 9180\,s + 9.09 \times 10^5}{s^2 + 1910\,s + 9.09 \times 10^5}$$

So:

$$H(s) = \frac{0.909\,s^2 + 9180\,s + 9.09 \times 10^5}{s^2 + 1910\,s + 9.09 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $0.909\,s^2 + 9180\,s + 9.09 \times 10^5$; divide by the leading coefficient 0.909 (this does not move the roots): $s^2 + 10100\,s + 10^6$; quadratic formula:

$$s = \frac{-10100 \pm \sqrt{(10100)^2 - 4(10^6)}}{2} = \frac{-10100 \pm 9900}{2} \Rightarrow s = -100,\ -10000$$

- Denominator: $s^2 + 1910\,s + 9.09 \times 10^5$; quadratic formula:

$$s = \frac{-1910 \pm \sqrt{(1910)^2 - 4(9.09 \times 10^5)}}{2} = \frac{-1910 \pm 90.9}{2} \Rightarrow s = -909,\ -1000$$

Factored form (leading coefficient in front):

$$H(s) = 0.909\,\frac{(s+100)(s+10000)}{(s+909)(s+1000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

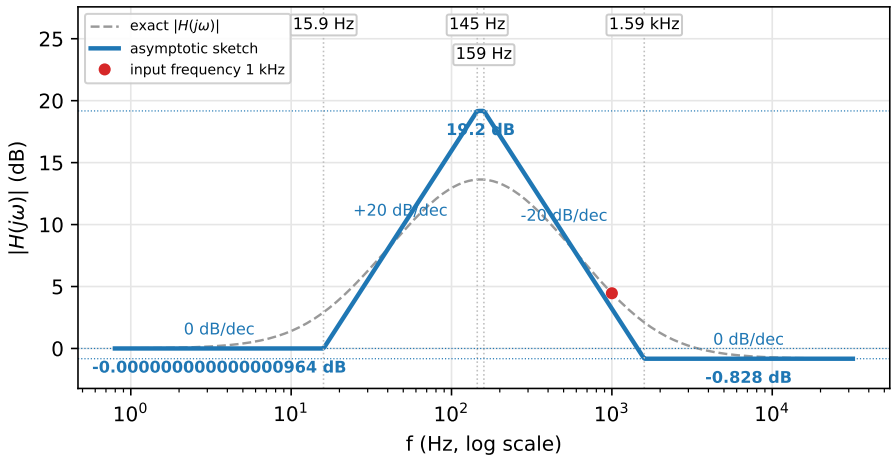
$$H(s) = 1\,\left(1 + \frac{s}{2\pi \cdot 15.9}\right)\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 145}}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 159}}\,\left(1 + \frac{s}{2\pi \cdot 1.59\text{k}}\right), \qquad K = 1$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
zero	100	15.9 Hz	+20 dB/dec
pole	909	145 Hz	−20 dB/dec
pole	1000	159 Hz	−20 dB/dec
zero	10000	1.59 kHz	+20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 1 = -0.000000000000000964$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 15.9 Hz	0 dB/dec (flat)	0 dB at end
15.9 Hz – 145 Hz	+20 dB/dec	0 dB at start, 19.2 dB at end
145 Hz – 159 Hz	0 dB/dec (flat)	19.2 dB at start, 19.2 dB at end
159 Hz – 1.59 kHz	−20 dB/dec	19.2 dB at start, -0.828 dB at end
$f >$ 1.59 kHz	0 dB/dec (flat)	-0.828 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin\left(2\pi\cdot 1000\,t + \frac{\pi}{4}\right)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
1	1	0
$\left(1+\dfrac{s}{2\pi\cdot 15.9}\right)$	62.84	1.55
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 145}}$	0.1432	-1.43
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 159}}$	0.1572	-1.41
$\left(1+\dfrac{s}{2\pi\cdot 1.59\text{k}}\right)$	1.181	0.561
total	**1.67** (product)	**−0.724** (sum)

$$|H(j\omega_{in})| = 1.67, \qquad \angle H(j\omega_{in}) = -0.724 \text{ rad}$$
$$v_{o,ac}(t) = 1.67 \times 0.1 \sin\left(2\pi\cdot 1000\,t + 0.0612\right) = 0.167 \sin\left(2\pi\cdot 1000\,t + 0.0612\right)$$

(phase: $\theta + \angle H = \frac{\pi}{4} + (-0.724) = 0.0612\,\text{rad}$.)

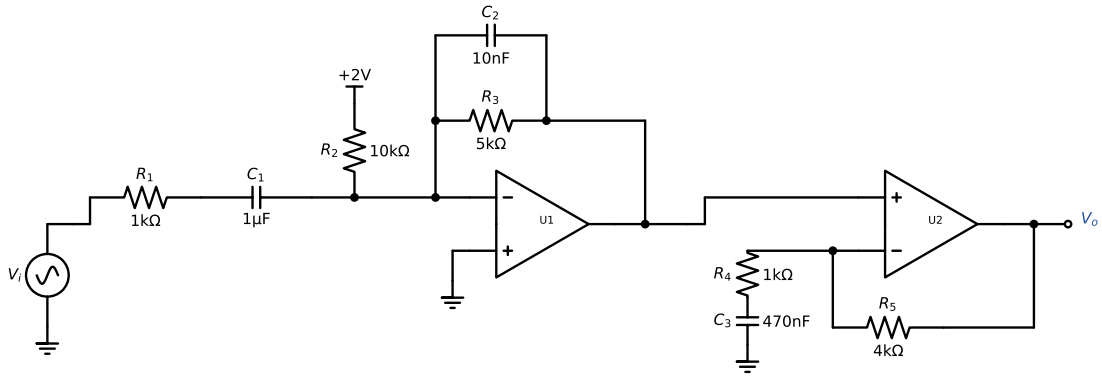
$$\boxed{V_o(t) = 0.167 \sin\left(2\pi\cdot 1000\,t + 0.0612\right)\ \text{V}}$$

Answer

$H(s) = 1\left(1 + \dfrac{s}{2\pi \cdot 15.9}\right)\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 145}}\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 159}}\left(1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}\right)$ (corners: 15.9 Hz, 145 Hz, 159 Hz, 1.59 kHz)

$V_o(t) = 0.167 \sin\left(2\pi \cdot 1000\, t + 0.0612\right)$ V

Solution 2A.15 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$ and the DC supply +2 V. Both DC and AC are present, so use superposition: $V_o(t) = V_{o,dc} + v_{o,ac}(t)$. The DC part is Step 2, the AC part Steps 3–7.

Step 2 — DC circuit $\Rightarrow V_{o,dc}$

DC rules: $V_i = 0$ and every capacitor is an open circuit (no inductors appear in these circuits). A block passes DC with its gain at $s = 0$, i.e. $H_k(0)$.

1. Block 1 (summer): Z_{in1} contains a capacitor, so it is open and carries no DC current. Only the supply drives a DC current, $\frac{V_{DC}}{R_2}$, through Z_{f1} (at DC $R_2 = 10\,\text{k}\Omega$, $Z_{f1} = 5\,\text{k}\Omega$):

$$V_{o1}\big|_{dc} = -Z_{f1}\frac{V_{DC}}{R_2} = -5\,\text{k}\cdot\frac{2}{10\,\text{k}} = -1\,\text{V}$$

2. Block 2: $H_2(0) = 1$ (capacitors open), so $V_o\big|_{dc} = 1 \times (-1) = -1\,\text{V}$.

$$\boxed{V_{o,dc} = -1\,\text{V}}$$

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its − input), so with an ideal op-amp (tool T1): (a) no current flows into the + or − input; (b) $V_- = V_+$ (virtual short). The circuit is a cascade of standard blocks, so each block is solved with its block formula (tool T2).

Step 4 — Transfer function, block by block

Each op-amp output is an ideal voltage source, so a block's output does not change when the next block is connected: the blocks can be solved one at a time and multiplied.

$$H(s) = \frac{V_o}{V_i} = \frac{V_{o1}}{V_i}\cdot\frac{V_o}{V_{o1}}$$

Block 1 (U1). Summing block: $V_- = V_+ = 0$ (virtual ground), so the input currents $\frac{V_i}{Z_{in1}}$ and $\frac{V_{DC}}{R_2}$ simply add and both flow through Z_{f1}: $V_{o1} = -Z_{f1}\left(\frac{V_i}{Z_{in1}} + \frac{V_{DC}}{R_2}\right)$. For the AC part the supply is zero, so R_2 (from virtual ground to AC ground) carries no AC current and drops out.

$$Z_{in1} = R_1 + \frac{1}{sC_1} = \frac{1000\,s + 10^6}{s}$$

$$Z_{f1} = R_3 \parallel \frac{1}{sC_2} = \frac{10^8}{s + 20000}$$

$$\frac{V_{o1}}{V_i} = -\frac{Z_{f1}}{Z_{in1}} = \frac{-10^5\, s}{s^2 + 21000\, s + 2 \times 10^7}$$

Block 2 (U2). Non-inverting block: The + input is the block input: $V_+ = V_{o1}$. The op-amp forces $V_- = V_+$, and Z_{g2}, R_5 divide the output: $V_- = \frac{Z_{g2}}{Z_{g2}+R_5} V_o$, hence $V_o = \left(1 + \frac{R_5}{Z_{g2}}\right) V_+$.

$$Z_{g2} = R_4 + \frac{1}{sC_3} = \frac{1000\, s + 2.13 \times 10^6}{s}$$

$$\frac{V_o}{V_{o1}} = \left(1 + \frac{R_5}{Z_{g2}}\right) = \frac{5\, s + 2130}{s + 2130}$$

Multiply the blocks and simplify:

$$H(s) = \frac{-5 \times 10^5\, s^2 - 2.13 \times 10^8\, s}{s^3 + 23100\, s^2 + 6.47 \times 10^7\, s + 4.26 \times 10^{10}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $-5 \times 10^5\, s - 2.13 \times 10^8$; divide by the leading coefficient -5×10^5 (this does not move the roots): $s + 426$; $s + 426 = 0 \Rightarrow s = -426$.
- Denominator: $s^3 + 23100\, s^2 + 6.47 \times 10^7\, s + 4.26 \times 10^{10}$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -1000,\ -2130,\ -20000$.

Factored form (leading coefficient in front):

$$H(s) = -5 \times 10^5\, \frac{s(s + 426)}{(s + 1000)(s + 2130)(s + 20000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

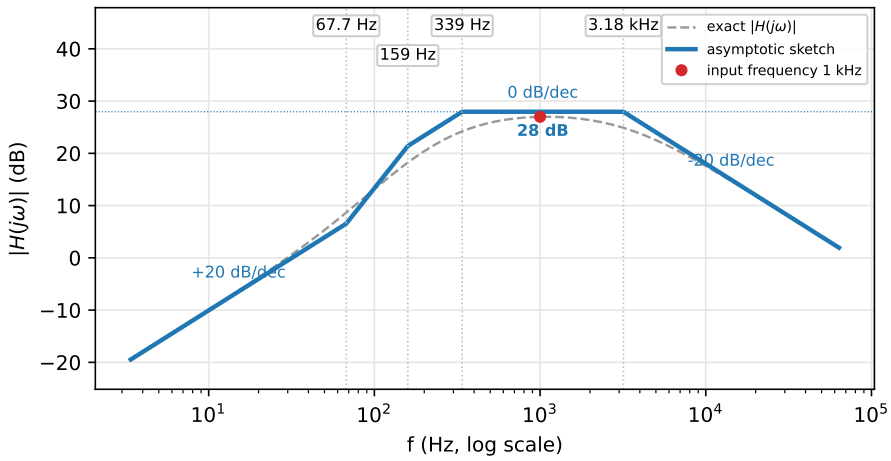
$$H(s) = -5\, \frac{s}{s + 2\pi \cdot 159} \left(1 + \frac{s}{2\pi \cdot 67.7}\right) \frac{1}{1 + \frac{s}{2\pi \cdot 339}}\, \frac{1}{1 + \frac{s}{2\pi \cdot 3.18\text{k}}}, \qquad K = -5$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
zero	426	67.7 Hz	+20 dB/dec
pole	1000	159 Hz	−20 dB/dec
pole	2130	339 Hz	−20 dB/dec
pole	20000	3.18 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.005\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 67.7$ Hz	+20 dB/dec	6.56 dB at end
67.7 Hz – 159 Hz	+40 dB/dec	6.56 dB at start, 21.4 dB at end
159 Hz – 339 Hz	+20 dB/dec	21.4 dB at start, 28 dB at end
339 Hz – 3.18 kHz	0 dB/dec (flat)	28 dB at start, 28 dB at end
$f > 3.18$ kHz	−20 dB/dec	28 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-5	5	-3.14
$\dfrac{s}{s+2\pi\cdot 159}$	0.9876	0.158
$\left(1+\dfrac{s}{2\pi\cdot 67.7}\right)$	14.8	1.5
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 339}}$	0.3207	-1.24
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 3.18\text{k}}}$	0.954	-0.304
total	**22.36** (product)	$-$**3.03** (sum)

$$|H(j\omega_{in})| = 22.4, \qquad \angle H(j\omega_{in}) = -3.03 \text{ rad}$$
$$v_{o,ac}(t) = 22.4 \times 0.05 \sin(2\pi\cdot 1000\,t - 3.03) = 1.12 \sin(2\pi\cdot 1000\,t - 3.03)$$

Add the DC part from Step 2 (superposition):

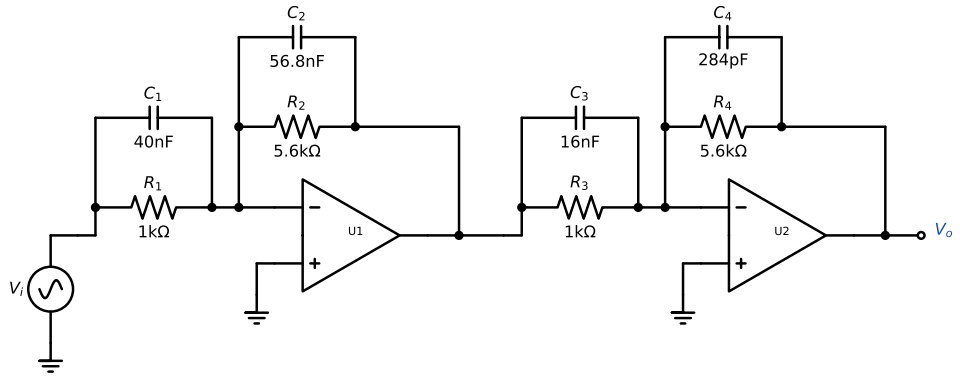
$$\boxed{V_o(t) = -1 + 1.12 \sin(2\pi\cdot 1000\,t - 3.03)\ \text{V}}$$

Answer

$H(s) = -5\,\dfrac{s}{s+2\pi\cdot 159}\left(1+\dfrac{s}{2\pi\cdot 67.7}\right)\dfrac{1}{1+\dfrac{s}{2\pi\cdot 339}}\dfrac{1}{1+\dfrac{s}{2\pi\cdot 3.18\text{k}}}$ (corners: 67.7 Hz, 159 Hz, 339 Hz, 3.18 kHz)

$V_o(t) = -1 + 1.12 \sin(2\pi\cdot 1000\,t - 3.03)$ V

Solution 2A.16 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a cascade of standard blocks, so each block is solved with its block formula (tool T2).

Step 4 — Transfer function, block by block

Each op-amp output is an ideal voltage source, so a block's output does not change when the next block is connected: the blocks can be solved one at a time and multiplied.

$$H(s) = \frac{V_o}{V_i} = \frac{V_{o1}}{V_i} \cdot \frac{V_o}{V_{o1}}$$

Block 1 (U1). Inverting block: the $+$ input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{Z_{in1}}$ through Z_{in1} cannot enter the op-amp, so all of it flows through Z_{f1}: $V_{o1} = 0 - Z_{f1} \cdot \frac{V_i}{Z_{in1}}$.

$$Z_{in1} = R_1 \parallel \frac{1}{sC_1} = \frac{2.5 \times 10^7}{s + 25000}$$

$$Z_{f1} = R_2 \parallel \frac{1}{sC_2} = \frac{1.76 \times 10^7}{s + 3140}$$

$$\frac{V_{o1}}{V_i} = -\frac{Z_{f1}}{Z_{in1}} = \frac{-0.704\,s - 17600}{s + 3140}$$

Block 2 (U2). Inverting block: the $+$ input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1} - 0}{Z_{in2}}$ through Z_{in2} cannot enter the op-amp, so all of it flows through Z_{f2}: $V_o = 0 - Z_{f2} \cdot \frac{V_{o1}}{Z_{in2}}$.

$$Z_{in2} = R_3 \parallel \frac{1}{sC_3} = \frac{6.25 \times 10^7}{s + 62500}$$

$$Z_{f2} = R_4 \parallel \frac{1}{sC_4} = \frac{3.52 \times 10^9}{s + 6.29 \times 10^5}$$

$$\frac{V_o}{V_{o1}} = -\frac{Z_{f2}}{Z_{in2}} = \frac{-56.3\,s - 3.52\times10^6}{s + 6.29\times10^5}$$

Multiply the blocks and simplify:

$$H(s) = \frac{39.7\,s^2 + 3.47\times10^6\,s + 6.2\times10^{10}}{s^2 + 6.32\times10^5\,s + 1.98\times10^9}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $39.7\,s^2 + 3.47\times10^6\,s + 6.2\times10^{10}$; divide by the leading coefficient 39.7 (this does not move the roots): $s^2 + 87500\,s + 1.56\times10^9$; quadratic formula:

$$s = \frac{-87500 \pm \sqrt{(87500)^2 - 4(1.56\times10^9)}}{2} = \frac{-87500 \pm 37500}{2} \Rightarrow s = -25000,\ -62500$$

- Denominator: $s^2 + 6.32\times10^5\,s + 1.98\times10^9$; quadratic formula:

$$s = \frac{-6.32\times10^5 \pm \sqrt{(6.32\times10^5)^2 - 4(1.98\times10^9)}}{2} = \frac{-6.32\times10^5 \pm 6.26\times10^5}{2} \Rightarrow s = -3140,\ -6.29{\times}10^5$$

Factored form (leading coefficient in front):

$$H(s) = 39.7\,\frac{(s+25000)(s+62500)}{(s+3140)(s+6.29\times10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

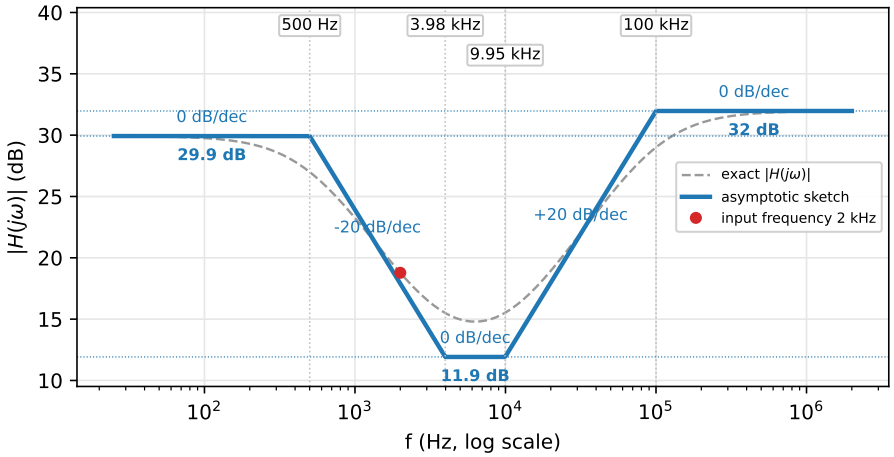
$$H(s) = 31.4\,\frac{1}{1+\dfrac{s}{2\pi\cdot 3.98\text{k}}}\left(1+\frac{s}{2\pi\cdot 3.98\text{k}}\right)\left(1+\frac{s}{2\pi\cdot 9.95\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}}, \qquad K = 31.4$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	3140	500 Hz	−20 dB/dec
zero	25000	3.98 kHz	+20 dB/dec
zero	62500	9.95 kHz	+20 dB/dec
pole	6.29×10^5	100 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 31.4 = 29.9$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 500$ Hz	0 dB/dec (flat)	29.9 dB at end
500 Hz – 3.98 kHz	−20 dB/dec	29.9 dB at start, 11.9 dB at end
3.98 kHz – 9.95 kHz	0 dB/dec (flat)	11.9 dB at start, 11.9 dB at end
9.95 kHz – 100 kHz	+20 dB/dec	11.9 dB at start, 32 dB at end
$f > 100$ kHz	0 dB/dec (flat)	32 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.01\sin(2\pi \cdot 2000\,t)$, so $\omega_{in} = 2\pi \cdot 2000 = 12600$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
31.4	31.36	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 500}}$	0.2427	-1.33
$\left(1+\dfrac{s}{2\pi\cdot 3.98\text{k}}\right)$	1.119	0.466
$\left(1+\dfrac{s}{2\pi\cdot 9.95\text{k}}\right)$	1.02	0.198
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}}$	0.9998	-0.02
total	**8.687** (product)	$-$**0.681** (sum)

$$|H(j\omega_{in})| = 8.69, \qquad \angle H(j\omega_{in}) = -0.681 \text{ rad}$$
$$v_{o,ac}(t) = 8.69 \times 0.01 \sin(2\pi \cdot 2000\,t - 0.681) = 0.0869 \sin(2\pi \cdot 2000\,t - 0.681)$$

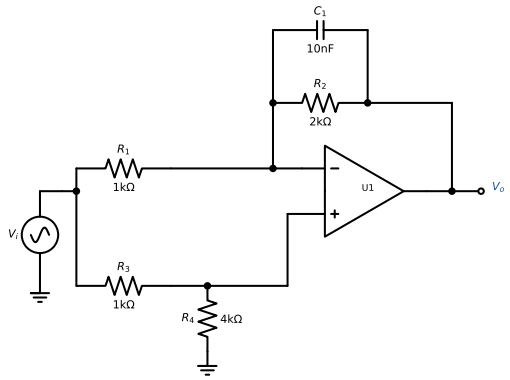
$$\boxed{V_o(t) = 0.0869 \sin(2\pi \cdot 2000\,t - 0.681) \text{ V}}$$

Answer

$$H(s) = 31.4\,\frac{1}{1+\dfrac{s}{2\pi\cdot 500}}\left(1+\frac{s}{2\pi\cdot 3.98\text{k}}\right)\left(1+\frac{s}{2\pi\cdot 9.95\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 100\text{k}}}$$ (corners: 500 Hz, 3.98 kHz, 9.95 kHz, 100 kHz)

$$V_o(t) = 0.0869\,\sin\left(2\pi\cdot 2000\,t - 0.681\right)\ \text{V}$$

Solution 2A.17 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

The op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a standard block, so use its block formula (tool T2).

Step 4 — Transfer function, block by block

Block 1 (U1). Difference block (both inputs driven by V_i): use superposition. Through the $+$ path the signal is divided, $V_+ = \frac{R_4}{R_3+R_4}V_i$, and amplified by the non-inverting gain $1 + \frac{Z_{f1}}{R_1}$; through the $-$ path it is amplified by the inverting gain $-\frac{Z_{f1}}{R_1}$. Add the two contributions.

$$Z_{f1} = R_2 \parallel \frac{1}{sC_1} = \frac{10^8}{s + 50000}$$

$$\frac{V_o}{V_i} = \frac{R_4}{R_3 + R_4}\left(1 + \frac{Z_{f1}}{R_1}\right) - \frac{Z_{f1}}{R_1} = \frac{0.8\,s + 20000}{s + 50000}$$

So:

$$H(s) = \frac{0.8\,s + 20000}{s + 50000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $0.8\,s + 20000$; divide by the leading coefficient 0.8 (this does not move the roots): $s + 25000$; $s + 25000 = 0 \Rightarrow s = -25000$.
- Denominator: $s + 50000$; $s + 50000 = 0 \Rightarrow s = -50000$.

Factored form (leading coefficient in front):

$$H(s) = 0.8\,\frac{(s + 25000)}{(s + 50000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1+\frac{s}{p}\right)$ and $(s+z) = z\left(1+\frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

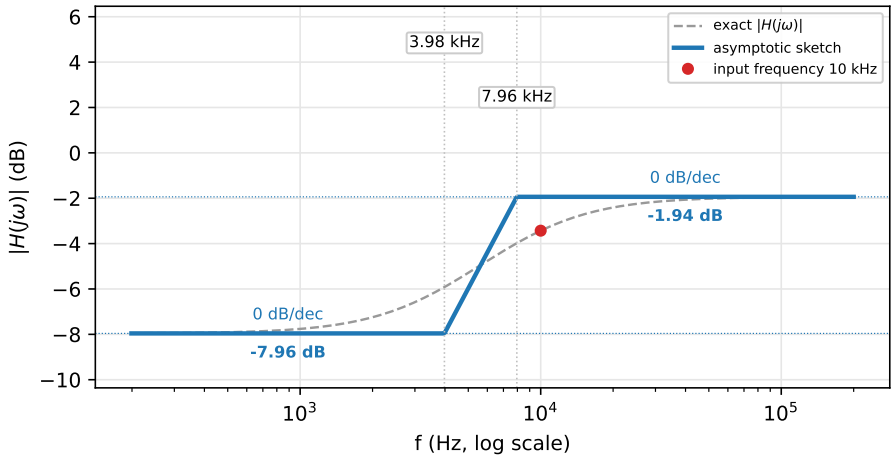
$$H(s) = 0.4\left(1+\frac{s}{2\pi\cdot 3.98\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 7.96\text{k}}}, \qquad K = 0.4$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
zero	25000	3.98 kHz	+20 dB/dec
pole	50000	7.96 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.4 = -7.96\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 3.98 kHz	0 dB/dec (flat)	-7.96 dB at end
3.98 kHz – 7.96 kHz	+20 dB/dec	-7.96 dB at start, -1.94 dB at end
$f >$ 7.96 kHz	0 dB/dec (flat)	-1.94 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\cos\left(2\pi\cdot 10000\,t - \frac{\pi}{3}\right)$, so $\omega_{in} = 2\pi\cdot 10000 = 62800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
0.4	0.4	0
$\left(1+\dfrac{s}{2\pi\cdot 3.98\text{k}}\right)$	2.705	1.19
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 7.96\text{k}}}$	0.6227	-0.899
total	**0.6737** (product)	**0.293** (sum)

$$|H(j\omega_{in})| = 0.674, \qquad \angle H(j\omega_{in}) = 0.293 \text{ rad}$$
$$v_{o,ac}(t) = 0.674 \times 0.5 \cos\left(2\pi\cdot 10000\,t - 0.754\right) = 0.337 \cos\left(2\pi\cdot 10000\,t - 0.754\right)$$

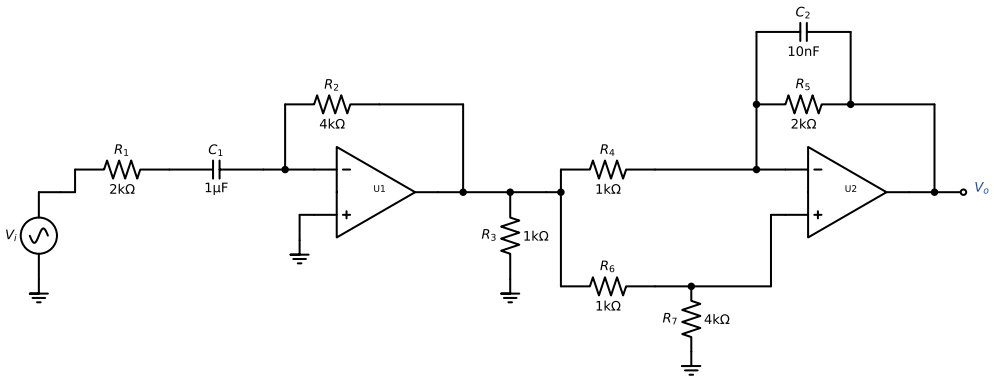
(phase: $\theta + \angle H = -\frac{\pi}{3} + (0.293) = -0.754\,\text{rad.}$)

$$\boxed{V_o(t) = 0.337 \cos\left(2\pi\cdot 10000\,t - 0.754\right)\ \text{V}}$$

Answer

$$H(s) = 0.4\left(1+\frac{s}{2\pi\cdot 3.98\text{k}}\right)\frac{1}{1+\dfrac{s}{2\pi\cdot 7.96\text{k}}} \quad \text{(corners: 3.98 kHz, 7.96 kHz)}$$
$$V_o(t) = 0.337 \cos\left(2\pi\cdot 10000\,t - 0.754\right)\ \text{V}$$

Solution 2A.18 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. No DC source: $V_{o,dc} = 0$ and Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — Op-amp rules

Each op-amp has negative feedback (its output is connected back to its $-$ input), so with an ideal op-amp (tool T1): (a) no current flows into the $+$ or $-$ input; (b) $V_- = V_+$ (virtual short). The circuit is a cascade of standard blocks, so each block is solved with its block formula (tool T2).

Step 4 — Transfer function, block by block

Each op-amp output is an ideal voltage source, so a block's output does not change when the next block is connected: the blocks can be solved one at a time and multiplied.

$$H(s) = \frac{V_o}{V_i} = \frac{V_{o1}}{V_i} \cdot \frac{V_o}{V_{o1}}$$

The load R_3 hangs on an op-amp output (an ideal voltage source): it draws current but does not change the output voltage, so it drops out of $H(s)$. **Block 1 (U1).** Inverting block: the $+$ input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{Z_{in1}}$ through Z_{in1} cannot enter the op-amp, so all of it flows through R_2: $V_{o1} = 0 - R_2 \cdot \frac{V_i}{Z_{in1}}$.

$$Z_{in1} = R_1 + \frac{1}{sC_1} = \frac{2000\,s + 10^6}{s}$$

$$\frac{V_{o1}}{V_i} = -\frac{R_2}{Z_{in1}} = \frac{-2\,s}{s + 500}$$

Block 2 (U2). Difference block (both inputs driven by V_{o1}): use superposition. Through the $+$ path the signal is divided, $V_+ = \frac{R_7}{R_6 + R_7} V_{o1}$, and amplified by the non-inverting gain $1 + \frac{Z_{f2}}{R_4}$; through the $-$ path it is amplified by the inverting gain $-\frac{Z_{f2}}{R_4}$. Add the two contributions.

$$Z_{f2} = R_5 \parallel \frac{1}{sC_2} = \frac{10^8}{s + 50000}$$

$$\frac{V_o}{V_{o1}} = \frac{R_7}{R_6 + R_7}\left(1 + \frac{Z_{f2}}{R_4}\right) - \frac{Z_{f2}}{R_4} = \frac{0.8\,s + 20000}{s + 50000}$$

Multiply the blocks and simplify:

$$H(s) = \frac{-1.6\,s^2 - 40000\,s}{s^2 + 50500\,s + 2.5 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $-1.6\,s - 40000$; divide by the leading coefficient -1.6 (this does not move the roots): $s + 25000$; $s + 25000 = 0 \Rightarrow s = -25000$.
- Denominator: $s^2 + 50500\,s + 2.5 \times 10^7$; quadratic formula:

$$s = \frac{-50500 \pm \sqrt{(50500)^2 - 4(2.5 \times 10^7)}}{2} = \frac{-50500 \pm 49500}{2} \Rightarrow s = -500,\ -50000$$

Factored form (leading coefficient in front):

$$H(s) = -1.6\,\frac{s(s + 25000)}{(s + 500)(s + 50000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

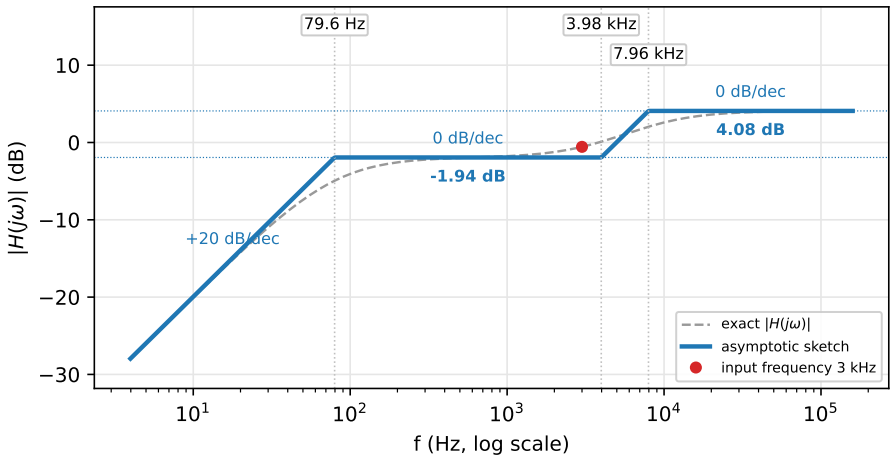
$$H(s) = -0.8\,\frac{s}{s + 2\pi \cdot 79.6}\left(1 + \frac{s}{2\pi \cdot 3.98\text{k}}\right)\frac{1}{1 + \dfrac{s}{2\pi \cdot 7.96\text{k}}}, \qquad K = -0.8$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	500	79.6 Hz	−20 dB/dec
zero	25000	3.98 kHz	+20 dB/dec
pole	50000	7.96 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.0016\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 79.6 Hz	+20 dB/dec	-1.94 dB at end
79.6 Hz – 3.98 kHz	0 dB/dec (flat)	-1.94 dB at start, -1.94 dB at end
3.98 kHz – 7.96 kHz	+20 dB/dec	-1.94 dB at start, 4.08 dB at end
$f >$ 7.96 kHz	0 dB/dec (flat)	4.08 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin\left(2\pi\cdot 3000\,t + \frac{\pi}{6}\right)$, so $\omega_{in} = 2\pi\cdot 3000 = 18800\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-0.8	0.8	-3.14
$\dfrac{s}{s+2\pi\cdot 79.6}$	0.9996	0.0265
$\left(1+\dfrac{s}{2\pi\cdot 3.98\text{k}}\right)$	1.252	0.646
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 7.96\text{k}}}$	0.9357	-0.361
total	**0.9372** (product)	$-$**2.83** (sum)

$$|H(j\omega_{in})| = 0.937, \qquad \angle H(j\omega_{in}) = -2.83 \text{ rad}$$
$$v_{o,ac}(t) = 0.937\times 0.1\sin\left(2\pi\cdot 3000\,t - 2.31\right) = 0.0937\sin\left(2\pi\cdot 3000\,t - 2.31\right)$$

(phase: $\theta + \angle H = \frac{\pi}{6} + (-2.83) = -2.31\,\text{rad}$.)

$$\boxed{V_o(t) = 0.0937\sin\left(2\pi\cdot 3000\,t - 2.31\right)\ \text{V}}$$

Answer

$$H(s) = -0.8\,\frac{s}{s + 2\pi \cdot 79.6}\left(1 + \frac{s}{2\pi \cdot 3.98\text{k}}\right)\frac{1}{1 + \dfrac{s}{2\pi \cdot 7.96\text{k}}} \quad \text{(corners: 79.6 Hz, 3.98 kHz, 7.96 kHz)}$$

$$V_o(t) = 0.0937\,\sin\left(2\pi \cdot 3000\,t - 2.31\right)\ \text{V}$$